

Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Administration

General Specifications

Customer Name

BaNES Council

Intersection/
General Description

Lower Bristol Road / Oak Street Puffin

Controller

☒ New

☐ Modification

Area Specifications/
Customer Drawings

Specification Section

Contract/Tender Ref:

Quotation No.

Works Order No.

Customer Order No.

Controller/
Serial Number

S.T.S. /EM Number

1084

Issue

1

Equipment
Installation by

Slot Cutting by

Civil Works by

Customer's Engineer

Telephone Number

Signal Company Use Only

Signal Engineer

(IF PROM Label as >)

PROM Number

766

PROM Variant

21

Configuration Check Value

EA 5F 36 AF

Controller Options

Hardware

ST750 ELV

Firmware Type and Issue

PB801 ISS 7

Other Options

ST950/ST900/ST750 Series Cabinet Options

Cabinet/Rack

Cabinet

Kit Type Options

☐

☐

☐ Small Non-UK

☒ Ped

Cabinet/Rack Variant

Cuckoo Options

☐

Mains Supply

230

Volts

50

Hz

Dimming

27.5 V

Answer Issue

12

Peak Lamp Current

1

Amps

Low Inrush Transformer

☐

Edit Issue

16

Average Lamp Power

1

Watts

Date Created

10/01/2023

Total Average Power

1

Watts

Power feed fuse rating: requires 30 Amp minimum for controller, 15 Amp minimum for pelican/lightly loaded controller

Phases, Stages and Streams

Phases, Stages and Streams

Add/Delete/Insert Streams:

Streams

Current Number of Streams

1

Stages

Current Number of stages
(inc. ALL-RED stages)

3

Phases

Current Total Number of Phases

3

Number of Real Phases

2

Number of Dummy Phases

1

Switched Signs

Number of Switched Signs

0

Action

Add At

Delete At

Facilities/Modes Enabled and Mode Priority Levels

Facilities

UTC

☒ Serial/Integral UTMC OTU

☐ Free-standing OTU

☐ Integral TC12 OTU

☒ Serial MOVA

☒ Master Time Clock

☐ Holiday Clock

☐

☐

☒ Lamp Monitoring

☒ RED Lamp Monitoring

☒ Pelican/Puffin/Toucan

☒ Standalone Manual

☐

☐ Speed Measurement

☐ Ripple Change

☐ London IMU

☐

☐ Non-UK

☐ Fail to Part Time

☐

☐ Last Lamp Failed Monitoring

☐

10

Starting Intergreen

Mode Priority

	1	2	3	4	5	6	7	8	9	10	11	12	13
☐ Part Time	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
☐ Selected FT or VA or CLF	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
☒ UTC	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
☐ CLF (Non-Base Time)	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
☒ CLF (Base Time)	☐	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
☒ Vehicle Actuated	☐	☐	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
☒ Fixed Time	☐	☐	☐	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐

Configuration Complexity

☐ Low

☐ Medium

☐ High

☒ Maximum

NoPromPB801.8.tif

Default PROM data file

Correspondence Monitoring to inc.

☒ Reds

☒ Ambers

☐ Switched Signs

☐

Flash Rate (ms)

400

Off

400

On

Works Order :
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Phases in Stages

Phases			
In Stages	A	B	C

Works Order :
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Stages in Streams

Stages in Streams

01234567

Phase or Stage to revert to in absence of demands/extensions

1

1

Startup Stage

Switch Off Stage

Standalone Pedestrian

Stages

012

0

Note: For a Stand-Alone Stream, the reversion must be to All Red stage or Traffic stage/phase to meet the relevant standard or specification.

Intersection : Lower Bristol Road / Oak Street Puffin

Phase Type and Conditions

1) App Types: 0 = Always Appears, 1 = Appears if dem'd prior to interstage, 2 = If dem'd, 3 = If dem'd before end of window time
2) Term Types: 0 = Term's at end of stage, 1 = Term's when Assoc phase gains R.O.W, 2 = Term's when Assoc phase loses R.O.W.
3) The HW Fail Flash fields are for information only on all but ST900 ELV and ST950 ELV Controllers. For other controllers, physical switches or links (etc.), select which aspects flash; these need to be set up manually.

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Opposing and Conflicting Phases

Select Stream(s) To Configure

☐ All

☐ 0

☐

☐

☐

☐

☐

☐

Initialse

☒ Amber Conflict Monitoring

		To Phase		
		A	B	C
From Phase	A		Co	
	B	Co		
	C			

• [REDACTED]

: 1084

████████████████████

Lower Bristol Road / Oak Street Puffin

Phase Minimums, Maximums, Extensions, Ped Leaving Periods

[illegible]

Note: For Standalone Streams see Help for use of Max Sets.

Works Order :
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Phase Intergreen Times

Select Stream(s) To Configure

☐ All ☐ 0 ☐ ☐ ☐ ☐ ☐ ☐ ☐

Note: On a Stand Alone Pelican/Toucan/Puffin Stream the Intergreens between Pedestrian and Traffic Phases are controlled by the timings (PBT, PIT, CMX, CDY, CRD and PAR), therefore 0 should be entered for the appropriate intergreen times in grid below.

To Phase		A	B	C
From Phase	A		3	
	B	5		
	C			

Works Order :
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Intergreen Handset Limits

HIGH

199

Copy Intergreen Values

		To Phase		
From Phase		A	B	C
	A		3	
	B	5		
	C			

Phase Timing Handset Ranges

Phase Timing Handset Ranges

Initialise Min Green Limits

Phase	Min. Green		Phase	Min. Green	
	Min.	Max.		Min.	Max.
A	7	63	Q		
B	6	63	R		
C	3	255	S		
D			T		
E			U		
F			V		
G			W		
H			X		
I			Y		
J			Z		
K			A2		
L			B2		
M			C2		
N			D2		
O			E2		
P			F2		

Max. Green

Min. 0Max. 255

Vehicle Extension

Min. 0.0Max. 10.0

Phase Delay

Min. 0Max. 0

Starting I/G

Min. 4Max. 12

Min Pedestrian Clearance (PBT)

Min. 1Max. 30

Traffic Phase Leaving

Min. 3.0Max. 4.0

Traffic Phase Red/Amber

Min. 2Max. 2

VA Demand and Extend Definitions

VA Demand and Extend Definitions

Demands

For Unlatched demands precede the name with a #.
Conditioning MUST be used to specify unlatched demands.

A	AIN1	AX2	AIN3	AX4
B	BPBU1	BPBU2	BPBU3	BPBU4
C				

Phases AtoP

Extensions

	AX2		AX4

Works Order :
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Phase Internal/Revertive Demands

Phase Internal/Revertive Demands

Start-up Vehicle Responsive Demands

A	☒	B	☒	C	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
	☐		☐		☐		☐		☐		☐		☐		☐	☐

Demands Inserted When Leaving Manual and Fixed Time Modes

A	☐	B	☐	C	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
	☐		☐		☐		☐		☐		☐		☐		☐	☐

Unlatched Demands that Start Max Timers

A	☒	B	☒	C	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
	☐		☐		☐		☐		☐		☐		☐		☐	☐

Revertive Phase Demands

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
Q	R	S	T	U	V	W	X	Y	Z	A2	B2	C2	D2	E2	F2

Phase - On Crossing and Kerbside Detector Definitions

On Crossing and Kerbside Input Definitions

Phase

A

B

C

On Crossing

BONC1	BONC3		

Phases AtoP

Kerbside

BKBS1	BKBS2	BKBS3	BKBS4

Works Order :
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Stream - Pelican/Puffin/Toucan Times

Stream - Pelican/Puffin/Toucan Times

Pedestrian Enable VA Mode (PEV)

Streams

01234567

1-Enabled

PedestrianAll RedTimes (Vehicle to Pedestrian)

Streams

01234567

(PAR n0) VA Gap Change3

(PAR n1) VA Max Change3

(PAR n2) FVP Change3

(PAR n3) UTC Change3

(PAR n4) Local Link Change3

Handset Range Limits

MinMax

13

Pelican Intergreen times

(PIT n0) Veh Red/Ped Flash Green0

(PIT n1) Veh Flash Amber/Ped Flash Green0

(PIT n2) Veh Flash Amber/Ped red0

(PIT n3) Veh Flash Amber/Ped Red Quiescent0

0000

0000

Works Order :
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Phase - Pelican, Puffin and Toucan Times

Phase - Pelican, Puffin and Toucan Times

Phase	PDD Ped Demand Delay	PDX Ped Demand Hold	CMX Clearance Maximum	CDY 0 Clearance Delay Gap Change	CDY 1 Clearance Delay Max Change	CRD Clearance Minimum Red
A	0	0.0	0	0	0	0
B	1	3.0	10	0	0	0
C	0	0.0	0	0	0	0

☒ Phases A to P

☐

☒ ST700/ST750 Audible/Tactile Drive Channel 1

☐

Pedestrian Handset Range Limits

	MIN	MAX
Demand Delay PDD	0	3
Demand Hold PDX	0.0	5.0
Clearance Maximum CMX	0	30
Clearance Delays CDY 0 and CDY1	0	3
Clearance Minimum Red CRD	0	3

Works Order : XXXXXXXXXX
EM Number : 1084
Engineer : XXXXXXXXXX
Intersection : Lower Bristol Road / Oak Street Puffin

IO and Link - Pelican/Puffin/Toucan Times

IO and Link - Pelican/Puffin/Toucan Times

Streams

01234567

Computer Control

PV

F1

Window Time
UIE

2

Local Link

PV1

Link Delay Time
LKD

0

Link Window Time
LKW

1

Link Override Time
LKO

100

Kerbside Mat Test
Output

Works Order :
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Pelican, Puffin, Toucan Pushbutton/Kerbside Associations

Pelican, Puffin, Toucan Pushbutton/Kerbside Associations

	Phase	Demand	KBS
0	B	BPBU1	BKBS1
1	B	BPBU2	BKBS2
2	B	BPBU3	BKBS3
3	B	BPBU4	BKBS4
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			

	Phase	Demand	KBS
16			
17			
18			
19			
20			
21			
22			
23			
24			
25			
26			
27			
28			
29			
30			
31			

	Phase	Demand	KBS
32			
33			
34			
35			
36			
37			
38			
39			
40			
41			
42			
43			
44			
45			
46			
47			

	Phase	Demand	KBS
48			
49			
50			
51			
52			
53			
54			
55			
56			
57			
58			
59			
60			
61			
62			
63			

Note: Any association pushed off the screen will have any previous association blanked.

Works Order :
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Stage Internal Demands/Pedestrian Window Times

Stage Internal Demands/Pedestrian Window Times

Start-up Vehicle Responsive Demands

0	☐	1	☐	2	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
	☐		☐		☐		☐		☐		☐		☐		☐	☐

Demands Inserted When Leaving Manual and Fixed Time Modes

0	☐	1	☐	2	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
	☐		☐		☐		☐		☐		☐		☐		☐	☐

Unlatched Demands that Start Maximum Timers

0	☒	1	☒	2	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
	☐		☐		☐		☐		☐		☐		☐		☐	☐

Window Times

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0	0													
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31

Exceptional Stages

☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Works Order :
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Fixed Time

Fixed Time

Stage Moves & Times (Not Fixed Time to Current Max)

Current Stage	0	1	2	3	4	5	6	7
Next Stage	1	2	1					
Time	0	0	0					
Current Stage	8	9	10	11	12	13	14	15
Next Stage								
Time								
Current Stage	16	17	18	19	20	21	22	23
Next Stage								
Time								
Current Stage	24	25	26	27	28	29	30	31
Next Stage								
Time								

Phases Demanded and Extended under Fixed Time to Current Max.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
Demand																
Extend																
	Q	R	S	T	U	V	W	X	Y	Z	A2	B2	C2	D2	E2	F2
Demand																
Extend																

Works Order :
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Intersection : Lower Bristol Road / Oak Street Puffin

CLF - Base Time

CLF - Base Time

Controller Base Date

XX/XX/XX

Controller Base Time

02:00:00

Plan Offset

	Minutes	Seconds		Minutes	Seconds
Plan 0	0	0	Plan 8	0	0
Plan 1	0	0	Plan 9	0	0
Plan 2	0	0	Plan 10	0	0
Plan 3	0	0	Plan 11	0	0
Plan 4	0	0	Plan 12	0	0
Plan 5	0	0	Plan 13	0	0
Plan 6	0	0	Plan 14	0	0
Plan 7	0	0	Plan 15	0	0

Handset Range Limits

	Minutes	Seconds
Min	0	0
Max	4	59

Works Order : XXXXXXXXXX
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Engineer : XXXXXXXXXX
Intersection : Lower Bristol Road / Oak Street Puffin

UTC General Data

UTC General Data

Type of UTC

☒ 106

☐ 316

Integral OTU Address

2

Number of Control Words

4

Number of Reply Words

☐

Controller to respond to TC bit.

☐

Introduction of UTC to be disabled by Priority and LRT M

Non UTC RTC synchronisation input name

RTC Synchronisation Times

Clock Synchronise Time (UTC TS input)

Day

Time

Time Only

02:00:00

Clock Confirm Time (UTC RT output)

Day

Time

Time Only

02:00:00

Works Order :
EM Number : 1084
Engineer :
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UTC Control and Reply Data Format

UTC Control and Reply Data Format

	Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Bit 8
Control Words								
Word 1	F1	#F2	D2	TS	MO			
Word 2								
Word 3								
Word 4								
Reply Words								
Word 1	G1	G2	DR2	CC	CF	DF	LF1	LF2
Word 2	RR1	MR	ML	MF				
Word 3								
Word 4								
Word 5								
Word 6								
Word 7								
Word 8								
Word 9								
Word 10								
Word 11								
Word 12								
Word 13								
Word 14								

Works Order :
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UTC Phase Demand and Extend Definitions

UTC Demand and Extend Definitions

Demands

For Unlatched demands, precede the name with a #.
Conditioning MUST be used to specify unlatched demands.

A				
B	D2			
C				

☒ Phases AtoP

☐

Extensions

UTC Stage and Mode Data Definitions

UTC Stage and Mode Data Definitions

Stage	Force Bit	Green Confirm Bit	Demand Confirm Bit	Stage	Force Bit	Green Confirm Bit	Demand Confirm Bit
0				16			
1	F1	G1		17			
2	#F2	G2	DR2	18			
3				19			
4				20			
5				21			
6				22			
7				23			
8				24			
9				25			
10				26			
11				27			
12				28			
13				29			
14				30			
15				31			

Mode Data Definitions

Manual Mode Operative:

☐ G1/G2

☐ RR

☐

Manual Mode Selected:

☐ G1/G2

☐ RR

☐

No Lamp Power, or Lamps Off due to RLM or Part Time:

☒ G1/G2

☐

☐

Detector Fault:

☐

☐

☒ DF

Normal NOT selected on the Manual Panel:

☐ G1/G2

☐ RR

☐

RR Button Selected:

☐ G1/G2

☐ RR

☐

If UTC Reply Confirms are required for a Controller Fault (CF) OR for separate MC and RR replies, Conditioning must be used.

UTC and MOVA Detectors

UTC and MOVA Detectors

Detector Mapping

☒ Combined

Set Selection

1

AIN1

2

AX2

3

AIN3

4

AX4

5

6

7

8

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Note - only 32 detectors available on MOVA 4.0

MTC - Time Switch Parameters

MTC - Time Switch Parameters

	Type	Event		Type	Event
0	Alternate Max	MAXSETB	16	No Action	
1	Alternate Max	MAXSETC	17	No Action	
2	Alternate Max	MAXSETD	18	No Action	
3	Alternate DFM	ALTDFMB	19	No Action	
4	Alternate DFM	ALTDFMC	20	No Action	
5	Alternate DFM	ALTDFMD	21	No Action	
6	Audibles	AUDMSK	22	No Action	
7	No Action		23	No Action	
8	No Action		24	No Action	
9	No Action		25	No Action	
10	No Action		26	No Action	
11	No Action		27	No Action	
12	No Action		28	No Action	
13	No Action		29	No Action	
14	No Action		30	No Action	
15	No Action		31	No Action	

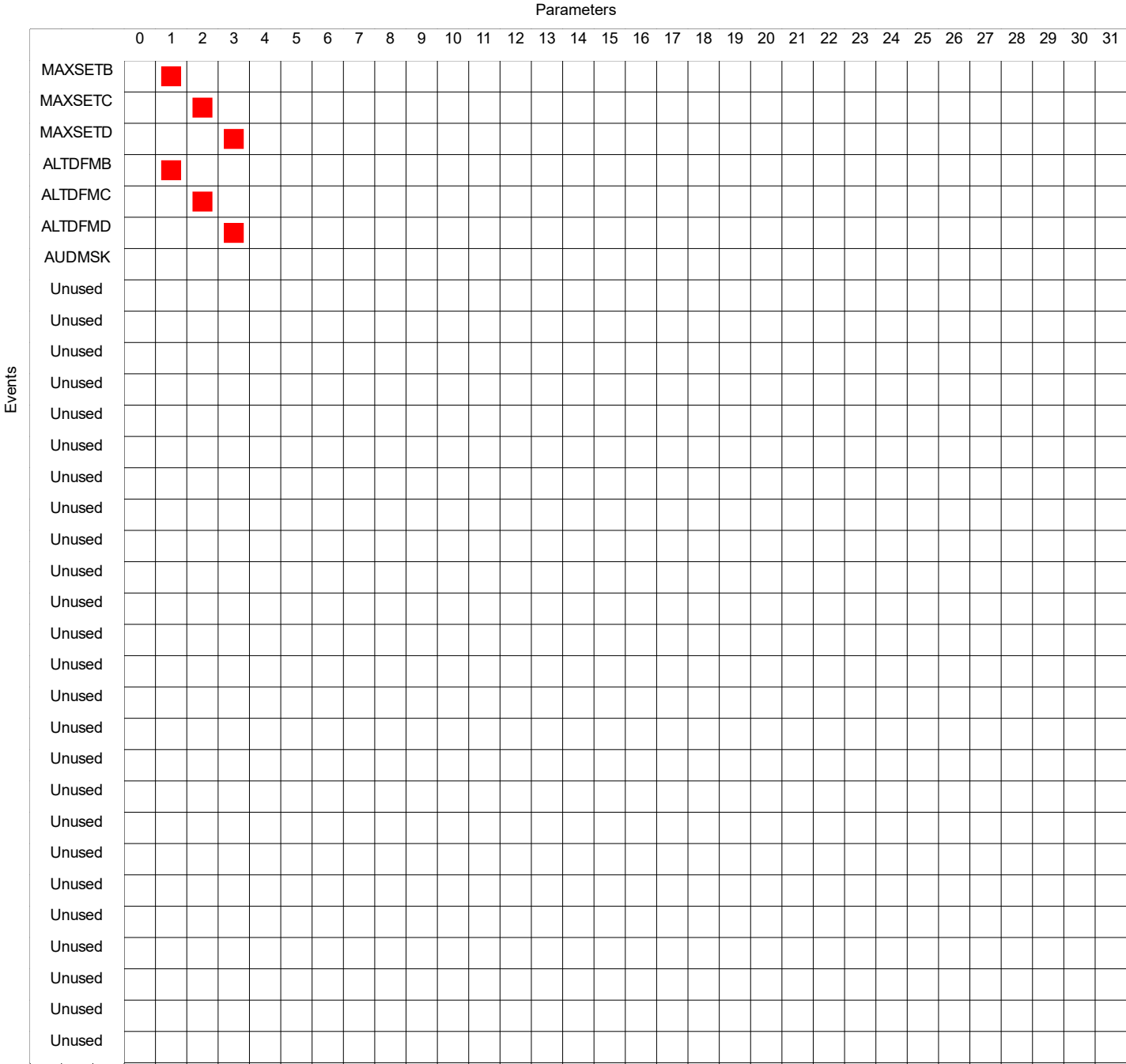
Works Order : XXXXXXXXXX

EM Number : 1084

Engineer : XXXXXXXXXX

Intersection : Lower Bristol Road / Oak Street Puffin

MTC - Time Switch Parameters Array



Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

MTC - Day Type

MTC - Day Type

No.	Mon	Tue	Wed	Thu	Fri	Sat	Sun
0	☐	☐	☐	☐	☐	☒	☐
1	☐	☐	☐	☐	☐	☐	☒
2	☒	☐	☐	☐	☐	☐	☐
3	☐	☒	☐	☐	☐	☐	☐
4	☐	☐	☒	☐	☐	☐	☐
5	☐	☐	☐	☒	☐	☐	☐
6	☐	☐	☐	☐	☒	☐	☐
7	☒	☒	☒	☒	☒	☒	☒
8	☒	☒	☒	☒	☒	☒	☐
9	☒	☒	☒	☒	☒	☐	☐
10	☐	☐	☐	☐	☐	☐	☐
11	☐	☐	☐	☐	☐	☐	☐
12	☐	☐	☐	☐	☐	☐	☐
13	☐	☐	☐	☐	☐	☐	☐
14	☐	☐	☐	☐	☐	☐	☐
15	☐	☐	☐	☐	☐	☐	☐

Works Order :
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MTC - Timetable

MTC - Timetable

View Timetable Settings

☒ 0 - 15

☐ 16 - 31

☐ 32 - 47

☐ 48 - 63

No.	Day Type	Time	Description	Function Code	Plan/Parameter
0	7	07:30:00	ENABLE AUDIOS	4	6
1	7	22:30:00	DISABLE AUDIOS	3	6
2	7	07:30:00	MAXSET A	2	0
3	7	09:30:00	MAXSET B	2	1
4	7	15:30:00	MAXSET C	2	2
5	7	19:00:00	MAXSET D	2	3
6	0			0	0
7	0			0	0
8	0			0	0
9	0			0	0
10	0			0	0
11	0			0	0
12	0			0	0
13	0			0	0
14	0			0	0
15	0			0	0

Function Codes:

0 = Isolate From CLF

1 = Introduce a CLF Plan

2 = Introduce a Parameter
(Combination of event switches)

3 = Selects an Individual event
switch to be set

4 = Selects an Individual event
switch to be cleared.

LMU - General

LMU - General

Lamp Monitoring - LMU Voltage

☐

☒ 48

☐

☐

Red Lamp Monitoring

Max Red Bulb Wattage

First Red Lamp Fault Speed

9

☐ RLF2 Cancels RLM additional Intergreens

RLM Additional Intergreen Handset Limits

Minimum

Maximum

3

10

☒ RLF2 Only Cleared by RFL = 1

☐ RLF1 Only Cleared by RFL = 1

Streams with Phase BlackOut on RLF2

☒ 0

☒ 1

☐

☐

☐

☐

☐

☐

Works Order :
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LMU Sensors (Built-in)

LMU Sensors (Built-in)

No. of LSLS cards fitted
1

HPU Connection
1

Sensor Configuration For LSLS 1 (Cabinet 1) (Reversed)

Phase	Aspect	Sensor #	Sensor Type	Phase	Aspect	Sensor #	Sensor Type
A	Red	1	As Seq.	N/A	N/A		
A	Red	2	As Seq.	N/A	N/A		
A	Red	N/A		N/A	N/A		
A	Red	N/A		N/A	N/A		
A	Red	N/A		N/A	N/A		
A	Amber	1	As Seq.	N/A	N/A		
A	Green	1	As Seq.	N/A	N/A		
B	Red	6	R,G	N/A	N/A		
B	Red	7	R,G	N/A	N/A		
B	Amber	8	Wait	N/A	N/A		
B	Amber	9	Wait	N/A	N/A		
B	Green	6	R,G	N/A	N/A		
B	Green	7	R,G	N/A	N/A		
B	Green	N/A		N/A	N/A		
B	Green	N/A		N/A	N/A		
B	Green	N/A		N/A	N/A		

Note : A (*) character next to a sensor number indicates that the sensor would also be available on the External sensors screen. Please be sure you wish to use these sensors here, as they will then become unavailable for Regulatory Signs.

LMU Sensors (External) for Regulatory Signs

LMU Sensors (External) for Regulatory Signs

External Sensors (1)

Sensor	Sensor Type
48	None
47	None
46	None
45	None

External Sensors (2)

Sensor	Sensor Type
--------	-------------

External Sensors (3)

Sensor	Sensor Type
--------	-------------

External Sensors (4)

Sensor	Sensor Type
--------	-------------

External Sensors (5)

Sensor	Sensor Type
--------	-------------

External Sensors (6)

Sensor	Sensor Type
--------	-------------

Note: Sensors which have been used as Onboard sensors will not be available here.

LMU Sensor Load Types

LMU Sensor Load Types

Page

1 of 1

Sensor	Phase	Sensor Type	LED R+W	RLM	Load Type	LLF Profile
1	A	As Seq.			1: Siemens Helios ELV	
2	A	As Seq.			1: Siemens Helios ELV	
6	B	R,G			3: Siemens LED Near Side Indicator	
7	B	R,G			3: Siemens LED Near Side Indicator	
8	B	Wait			2: Siemens LED Demand Indicator	
9	B	Wait			2: Siemens LED Demand Indicator	

Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

RLM Phase Inhibits

Phases Inhibited/Black-Out

Phases with RLF2	A	B	C
	A		
	B		
	C		

Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Manual Panel

Manual Panel

Stage Buttons and LEDs

Button No.	Title	Called Stage for Stream							
		0	1	2	3	4	5	6	7
0									
1									
2									
3									
4									
5									
6									
7									

General LEDs

	AUX 1	AUX 2	AUX 3	AUX 4 (Hurry Call)	AUX 5 (Higher Priority)
Conditioned	☒	☐	☐	☐	☐

General Buttons

	None	SW1	SW2	SW3
Momentary		☐	☐	☐
Dim Override	☒	☐	☐	☐
RR	☒	☐	☐	☐

Manual Mode Enable

☒
☐
☐

NOTE:
For this to operate Special
Conditioning is required.

Manual Signals On

☐
☒

Mode Select Switches Disabled

☐☐☐

Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Special Conditioning

```
; IN LOOPS TO EXTEND PHASE A IF REQUIRED ENABLED BY CFE13=1
; =====

(AIN1_EXT+AIN3_EXT)./CFE13:+=EXOAS ; AIN OR AIN3 DETECTORS TO EXTEND PHASE A UNLESS DISABLED BY CFE13=1
      *+=EXCA ; (DEFAULT STATE IS ACTIVE)

; PED WAIT DEMANDS TO MOVA
; =====

PRSLMPAB.NOT(LMUINHB)=MOVADET5 ; DEMAND FOR PED B SETS MOVADET 5 UNLESS INHIBITED BY RLM

; MANUAL PANEL LED'S
; ~~~~~
; AUX 1 LED LIT FOR UTC MODE
; =====
; AUX 1 LED FLASH WHEN MOVA MODE ACTIVE
; =====
IFT (MODE0 EQL<8>) THN ; START UP MODE HOLDS LED FLASH TIMER 2 RESET.
  RUN<2>
END
IFT CNDTER2 THN ; FLASH TIMER 2 EXPIRY STARTS FLASH TIMER 3.
  RUN<3>
END
IFT CNDTER3 THN ; FLASH TIMER 3 EXPIRY STARTS FLASH TIMER 2.
  RUN<2>
END

IFT ((MODE0 EQL<14>).(F1+F2+CNDTMA0)).NOT(MOVATO) THN
  TRUE=MIL22 ; AUX 1 LED ON WHEN UTC MODE ACTIVE
ELS
  FALSE=MIL22
  IFT ((MODE0 EQL<14>).(F1+F2).(MOVATO)) THN
    CNDTMA3=MIL22 ; AUX 1 LED FLASH WHEN MOVA MODE ACTIVE
  END
END

; CONTROL READY BIT - 1
; =====

; Higher Priority L.E.D. not used in this Ped config and AUX 3 L.E.D. to be enabled in conditioning
; PIR12 - Lamps off Delay Timer Set at 1.0 Limits 1.0 - 5.0
; PIR13 - Mova Override Timer Set at 10.0 limits 5.0 - 31.8
; PIR14 - CRB Delay Timer Set at 120 Limits 0 - 255

IFT (PRSLMPRA+PRSLMPAA+PRSLMPGA) THN ; TIMER TO STOP MOVACRB BEING TOGGLED BY THE LAMP TEST
  RUN<12>
END
IFT NOT(F1+F2).NOT(CNDTMA14).(MODE0 EQL<14>) THN ; NOT IN MOVA MODE RUN TIMER
  RUN<14> ; PIR14=120
END
IFT CNDTER14+(F1+F2) THN
  LOD<10> 1SCRTCH10
  TRUE=2SCRT8
END ; START A 2 SEC INTERNAL TIMER FOR CRB TOGGLE
/(1SCRTST10 EQL<0>)=.2SCRT8
IFT (1SCRTST10 GRT<0>) THN
  DEC 1SCRTCH10
END
(MODE0 EQL<14>).(NOT(2SCRT8)+(F1+F2)) $
.CNDTMA12=MOVACRB
```

Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Special Conditioning

```
; MOVA FORCE BITS TO HOLD PED STAGE
; =====

IFT (F1+F2).MOVATO THN                ; MOVA bits present, MOVA in control.
  F1: = PRVST2                        ; MOVAF1 - prevent the move to stage 2.
  * = +UCST1                          ; MOVAF1 - create a demand for stage 1.
  F2: = PRVST1                        ; MOVAF2 - prevent the move to stage 1.
  * = +UCST2                          ; MOVAF2 - create a demand for stage 1.
ELS
  FALSE = PRVST1                      ; normal vehicle extensions and drop
  FALSE = PRVST2                      ; stage prevents.
END

; UTC F1/F2
; =====
;
STAGE2 = +UCST1                        ; if we're in Stage2, demand Stage1 to revert to traffic.
;
F1=SCINHA                             ; if we're in UTC mode, inhibit vehicle extensions.
```

Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Special Conditioning

```
;UTC REPLY BITS  
;=====
```



```
NOT(FLF55) = LF1      ;Any Lamp failure  
LMPON.LPSPRD.RIMMSKA.NOT(FLF17)./STAGE1=G1  ; Lamps Supply Off REPLIES G1/G2  
LMPON.LPSPRD.RIMMSKA.NOT(FLF17)./STAGE2=G2  
NOT(FLF22) =LF2      ;1st/2nd Red Lamp Failure  
NOT(FLFCOM) = CF     ;CONTROLLER FAULT REPLY BIT  
/(MODE0 EQL<9>)=RR1  ; NOT IN NORMAL REPLIES RR1 - FIXED VEHICLE PERIOD
```

Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Special Conditioning

```
; AUDIBLE MONITORING
; =====

IFT CFE6.AUD_ML1_USED THN                                ; IF THE FACILITY IS ENABLED (ON PED1)
FALSE = AUDDIS0                                           ;   ENABLE THE AUDIBLE OUTPUTS
FDH18B0=+1CN1FLD0                                         ;   ENSURE RFL=1 CAN'T CLEAR FAULT WITHOUT POWER OFF/ON

IFT (AUD_ML1+AUD_MQ1) XOR (PRSLMPGB.PRSLMPRA.NOT(AUDDIS0+PRSFLSGB)) THN
FALSE = 1SCRT27                                           ;   IF AUDIBLES DONT MATCH REQUESTED STATE
IFT NOT(1SCRT25) THN                                     ;   RESET FLAG (TO RESTART 'GOOD' TIMER)
  RUN<5>                                                  ;   IF BAD TIMER NOT STARTED
  TRUE = 1SCRT25                                          ;   START BAD TIMER
ELSE                                                      ;   SET FLAG
  1SCRT25.NOT(CNDTMA5)=+1CN1FLD0                         ;   IF BAD TIMER EXPIRES SET FLD 18:00000001
END
ELSE                                                      ;   ELS (AUDIBLE OUTPUTS APPEAR OK)
IFT NOT(1SCRT27) THN                                     ;   IF GOOD TIMER NOT STARTED
  RUN<7>                                                  ;   START GOOD TIMER
  TRUE = 1SCRT27                                          ;   SET FLAG
ELSE                                                      ;   ELS
  CNDTMA7=.1SCRT25                                       ;   IF GOOD TIMER EXPIRES STOP BAD TIMER
END                                                       ;   (AUDIBLES HAVE BEEN 'GOOD' FOR A WHILE)
END
ELSE                                                      ; ELS FACILITY DISABLED
TRUE = AUDDIS0                                           ;   DISABLE THE AUDIBLE OUTPUTS
FALSE = 1CN1FLD0                                         ;   CLEAR FLD 18
FALSE = 1SCRT25                                          ;   RESET FLAG (TO RESTART TIMER)
FALSE = 1SCRT27                                          ;   RESET FLAG (TO RESTART TIMER)
END                                                       ; END OF 'IS FACILITY ENABLED'

IFT CFE6.AUD_ML2_USED THN                                ; IF THE FACILITY IS ENABLED (ON PED2)
FALSE = AUDDIS1                                           ;   ENABLE THE AUDIBLE OUTPUTS
FDH18B1=+1CN1FLD1                                         ;   ENSURE RFL=1 CAN'T CLEAR FAULT WITHOUT POWER OFF/ON

IFT (AUD_ML2+AUD_MQ2) XOR (PRSLMPGD.PRSLMPRC.NOT(AUDDIS1+PRSFLSGD)) THN
FALSE = 1SCRT28                                           ;   IF AUDIBLES DONT MATCH REQUESTED STATE
IFT NOT(1SCRT26) THN                                     ;   RESET FLAG (TO RESTART 'GOOD' TIMER)
  RUN<6>                                                  ;   IF BAD TIMER NOT STARTED
  TRUE = 1SCRT26                                          ;   START BAD TIMER
ELSE                                                      ;   SET FLAG
  1SCRT26.NOT(CNDTMA6)=+1CN1FLD1                         ;   IF BAD TIMER EXPIRES SET FLD 18:00000010
END
ELSE                                                      ;   ELS (AUDIBLE OUTPUTS APPEAR OK)
IFT NOT(1SCRT28) THN                                     ;   IF GOOD TIMER NOT STARTED
  RUN<8>                                                  ;   START GOOD TIMER
  TRUE = 1SCRT28                                          ;   SET FLAG
ELSE                                                      ;   ELS
  CNDTMA8=.1SCRT26                                       ;   IF GOOD TIMER EXPIRES STOP BAD TIMER
END                                                       ;   (AUDIBLES HAVE BEEN 'GOOD' FOR A WHILE)
END
ELSE                                                      ; ELS FACILITY DISABLED
TRUE = AUDDIS1                                           ;   DISABLE THE AUDIBLE OUTPUTS
FALSE = 1CN1FLD1                                         ;   CLEAR FLD 18
FALSE = 1SCRT26                                          ;   RESET FLAG (TO RESTART TIMER)
FALSE = 1SCRT28                                          ;   RESET FLAG (TO RESTART TIMER)
END                                                       ; END OF 'IS FACILITY ENABLED'

1CN1FLD=1CN1FLF                                           ; SET FLF 27:255 SCF1 IF ANY FAULT DATA BITS SET
```


Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Special Conditioning

1CN1FLF + FFH27 = SOFCND ; REQUEST SIGNALS OFF IF FLF 27 SET

Special Conditioning Timers

Special Conditioning Timers

Timers

0-31

No	Value	Min	Max	200ms	Description	No	Value	Min	Max	200ms	Description
0	2	0	255	☐	UTC 2s PV WINDOW TIMER	16		0	255	☐	
1		0	255	☐		17		0	255	☐	
2	1	1	3	☐	FLASH TIMER 1	18		0	255	☐	
3	1	1	3	☐	FLASH TIMER 2	19		0	255	☐	
4		0	255	☐		20		0	255	☐	
5	1.0	0.2	5.0	☒	Audible fault time (PED1)	21		0	255	☐	
6		0	255	☐		22		0	255	☐	
7	0.6	0.2	5.0	☒	Audible ok time (PED1)	23		0	255	☐	
8		0	255	☐		24		0	255	☐	
9		0	255	☐		25		0	255	☐	
10		0	255	☐		26		0	255	☐	
11		0	255	☐		27		0	255	☐	
12	1	1	255	☐	LAMPS OFF DELAY TIMER	28		0	255	☐	
13	10	5	255	☐	MOVA OVERRIDE TIMER	29		0	255	☐	
14	120	0	255	☐	CRB DELAY TIMER	30		0	255	☐	
15		0	255	☐		31		0	255	☐	

Fault Log Flags

Fault Log Flags

Fault No	Cond Flag	Act Flag
0	☒	☐
1	☒	☐
2	☒	☐
3	☒	☐
4	☒	☐
5	☒	☐
6	☒	☐
7	☒	☐
8	☒	☐
9	☒	☐
10	☒	☐
11	☒	☐
12	☐	☐
13	☒	☐
14	☒	☐
15	☒	☐

Fault No	Cond Flag	Act Flag
16	☒	☐
17	☐	☐
18	☒	☐
19	☒	☐
20	☒	☐
21	☒	☐
22	☐	☐
23	☒	☐
24	☒	☐
25	☒	☐
26	☒	☐
27	☒	☐
28	☒	☐
29	☒	☐
30	☒	☐
31	☒	☐

Fault No	Cond Flag	Act Flag
32	☒	☐
33	☒	☐
34	☒	☐
35	☒	☐
36	☒	☐
37	☒	☐
38	☒	☐
39	☒	☐
40	☒	☐
41	☒	☐
42	☒	☐
43	☒	☐
44	☒	☐
45	☒	☐
46	☒	☐
47	☒	☐

Fault No	Cond Flag	Act Flag
48	☒	☐
49	☒	☐
50	☒	☐
51	☒	☐
52	☒	☐
53	☒	☐
54	☒	☐
55	☐	☐
56	☒	☐
57	☒	☐
58	☒	☐
59	☒	☐
60	☒	☐
61	☒	☐
62	☒	☐
63	☒	☐

Note:

Cond Flag -
If a fault occurs which sets a fault log flag that has been checked for this Cond flag option then a flag will be set that can be read in Conditioning.

Act Flag -
If a fault occurs which sets a fault log flag that has been checked for this Act flag option then firstly the lamps

will be switched OFF and secondly a flag will be set that can be read in conditioning, to allow any further actions required to be performed by conditioning.

Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Special Instructions

Note For Printing This Configuration:

=====

Only Print the Administration Screen
and the Special Instructions.
The other screens should not be printed.

```
-----  
| ##### | ##### | ##### | ##### | ### | ##### | # | # | |
| # | # | # | # | # | # | # | # | # |  
| ### | # | # | # | # | # | # | # | # |  
| # | # | # | # | # | # | # | # | # |  
| ##### | # | # | # | # | # | # | # |  
-----  
| ST750 ELV SINGLE / DUAL NEAR SIDE CROSSING |  
-----
```

1.1) General Instructions

=====

These instructions must be read in conjunction with:
- the ST750ELV General Handbook (667/HB/32750/000), and
- the ST900/ST750 Family Handset Handbook (667/HH/32900/000 issue 4 or later).

This configuration provides the starting point for a ST750ELV single or dual pedestrian crossing with an extendable red clearance period for near-side ped signals, i.e. a near-side Puffin or Toucan Crossing. Once the configuration has been loaded into the controller, it will need to be modified using the handset before the signals are switched on.

Only the Administration screen and these special instructions should be printed with a paper copy placed in the controller. All controller values modified using the handset MUST be recorded in this document.

This configuration uses the following phase / stage / stream allocation:

Stream	Stage	Phase	Description
0	1	A	Vehicle Stage / Phase of the first stand-alone pedestrian stream
0	2	B	Pedestrian Stage / Phase of the first stand-alone pedestrian stream
1	3	C	Vehicle Stage / Phase of the second stand-alone pedestrian stream
1	4	D	Pedestrian Stage / Phase of the second stand-alone pedestrian stream

These special instructions are divided into the following sections:

- 1) General Instructions / Applicable Part Numbers / Installation
- 2) I/O Functions Provided / Provision Of Physical I/O Lines
- 3) Controller I/O Handset Commands
- 4) Lamp Monitoring Arrangement and Configuration Commands
- 5) Speed Discrimination/Assessment
- 6) Vehicle Timings
- 7) Pedestrian Timings
- 8) UTC / Local Link / Dimming / Manual Panel
- 9) Cableless Link Facility (CLF)
- 10) Controller Timetable

Configuration Complexity Number is 199

Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Special Instructions

Configuration Issue History

Issue	Released	By Whom	Changes
1	Dec 2008	Siemens	First release for Trial
2	Jan 2009	Siemens	0004522: Remove item lists because these may change. 0004528: Add this issue history.

If changes are required, contact Siemens Poole. Alternatively, take a copy of this configuration, assign it a new EM-number and make the required changes using similar processes to an intersection configuration. Do not modify this configuration without changing the EM-number. Only Siemens Poole should modify this standard configuration.

Differences between the ST700 and ST750ELV configurations

- 1) An ST750ELV uses PB801 firmware issue 7 or later; an ST700 uses PB800 firmware
- 2) Some I/O names and positions have changed; always refer to section 2
- 3) The inputs LLDEM1/2 can now provide the functionality of the BUS1/2 inputs, see section 6.1
- 4) Lamp faults and DFM faults set the CF bit if the LF and DF outputs are not used
- 5) Audibles are switched off/quiet between 11pm and 7:30am by default, see section 10.1
- 6) If pushbuttons and kerbside detectors are used, see section 2.1
- 7) Kerbside and On-Crossing Detector inputs are inverted by default (open for detect), see section 2.1

Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Special Instructions

1.2) Controller Connections

The following table shows the lamp drive outputs configured.

PED 1			PED 2		
Terminals	Description		Terminals		
RED1	1	Veh Phase - RLM Channel 1	RED1	17	
RED2	2	Veh Phase - RLM Channel 2	RED2	18	
RED3	3	Veh Phase - RLM Channel 3	RED3	19	
RED4	4	Veh Phase - RLM Channel 4	RED4	20	
RED5	5	Veh Phase - RLM Channel 5	RED5	21	
AMB	6	Veh Phase - Amber Signals	AMB	22	
GRN	7	Veh Phase - Green Signals	GRN	23	
REDM	8	Ped Phase - Red Signals	REDM	24	
REDM	9	Ped Phase - Red Signals	REDM	25	
DMND	10	Ped Phase - Demand Indicator/Wait	DMND	26	
DMND	11	Ped Phase - Demand Indicator/Wait	DMND	27	
GRNM	12	Ped Phase - Green Signals	GRNM	28	
GRNM	13	Ped Phase - Green Signals	GRNM	29	
TACT	14	Ped Phase - Tactile -48V Supply	TACT	30	> See section 1.3
TACT	15	Ped Phase - Tactile -48V Supply	TACT	31	
ADRV	16	Ped Phase - Audible Driver Supply	ADRV	32	

Check the Handbook for the maximum number of signals that can be connected to any one LSLS Output.
For Lamp Monitoring details, see section 4.

COM	QU	LO	1 \	AUD1 Audible Driver Outputs (See section 1.3)
COM	QU	LO	2 \	
COM	QU	LO	3 /	
COM	QU	LO	4 /	
MCOM	MQ	ML	5	AUD1 Audible Monitor Outputs (See section 1.3)
COM	LQ	EN	6	AUD2 Control signals (See section 1.3)
SOL+	SOL-	SIG	7	Solar Cell Supply (+/-) and Input (SIG)
NO1	COM	NC1	8 \	I/O Card 'Relay' Outputs (see section 2.2.1) Both Normally-Open and Normally-Closed contacts are provided. NO1/COM/NC1 = Output #24 NO2/COM/NC2 = Output #25 NO3/COM/NC3 = Otuput #26 NO4/COM/NC4 = Otuput #27
NO2	COM	NC2	9 \	
NO3	COM	NC3	10 /	
NO4	COM	NC4	11 /	

DET+	DET-	1 \	> Detector Supply (to AGDs and Detector Cards, etc.)
DET+	DET-	2	
DET+	DET-	3 /	
RET	T1	9 \	Torroid inputs to LSLS Card (from optional Reg Sign Kit)
RET	T2	10 /	
RET	+24V	11	Auxillary +24VDC (for internal use only, e.g. to London IMU)

Works Order :
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Intersection : Lower Bristol Road / Oak Street Puffin

Special Instructions

1.3) Audible / Tactile Connections

PED 1			PED 2		
TACT	14	Ped Phase - Tactile -48V Supply	TACT	30	
TACT	15	Ped Phase - Tactile -48V Supply	TACT	31	
ADRV	16*	Ped Phase - Audible Driver Supply	ADRV	32	

The LSLS Outputs labelled 'TACT' and 'ADRV' can be used to drive a number of Pedestrian Tactile Units. Check the Handbook for the maximum number of devices that can be connected to any one LSLS Output.
* The built-in Audible Driver is internally connected to Output #16 (ADRV)
This must be taken in to consideration when calculating the maximum load on this output.

The built-in ST750ELV Audible Driver is internally powered from ADRV-16.
This provides the following outputs on the ST750ELV backplane:

COM	QU	LO	1 \	AUD1 Audible Driver Outputs (for PED1).
COM	QU	LO	2 \	Connect each Audible Sounder between COM and either QU or LO.
COM	QU	LO	3 /	QU = Output ON during night (Quiet) } For the timetable,
COM	QU	LO	4 /	LO = Output ON during day (Loud) } refer to section 10.
MCOM	MQ	ML	5 -	AUD1 Audible Monitor Outputs (connect to I/O Inputs AUD_MQ1 and AUD_ML1)
COM	LQ	EN	6 -	AUD2 Control signals to optional Audible Driver Kit (PED2 - see below)

PED1 - First / Single Stream

In order to enable and monitor the AUD1 output drives to the audibles, follow these steps:

- 1) Connect input return to 'MCOM' on the ST750ELV backplane
- 2) Connect the input AUD_ML1 to 'ML' on the ST750ELV backplane
- 3) Connect the input AUD MQ1 to 'MQ' on the ST750ELV backplane
(Connect AUD_MQ1 even if only 'loud' / day-time audibles are fitted)
- 4) Allocate the inputs AUD_ML1 and AUD_MQ1 using IOA
- 5) Enable the audibles by entering CFE 6=1.
*** WITHOUT SETTING THE CFE COMMAND THE AUDIBLES WILL NOT FUNCTION ***

Works Order :
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Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Special Instructions

2.1.1) I/O Functions Provided (Detection & Misc I/O)

N	Mnemonic	In/Out	Def DFM	New DFM	Def IPS	New IPS	Def IPX	New IPX	Def IOA	New IOA	Term. Block	Pel	Description
0		I	Y 0		0		0.0		0			1	
1		I	Y 0		0		0.0		1			1	
2		I	Y 0		0		0.0		255			1	
3		I	Y 0		0		0.0		255			1	
4		I	Y 3		1		0.6		4			1	
5		I	Y 3		1		0.6		5			1	
6		I	Y 3		1		0.6		255			1	
7		I	Y 3		1		0.6		255			1	
8		I	Y 1		0		2.0		8			1	Pedestrian Pushbutton / Cycle Detector 1
9		I	Y 2		1		2.0		9			1	Pedestrian Kerbside detector 1 (*)
10		I	Y 1		0		2.0		10			1	Pedestrian Pushbutton / Cycle Detector 2
11		I	Y 2		1		2.0		11			1	Pedestrian Kerbside detector 2 (*)
12		I	Y 1		0		2.0		255			1	Pedestrian Pushbutton / Cycle Detector 3
13		I	Y 2		1		2.0		255			1	Pedestrian Kerbside detector 3 (*)
14		I	Y 1		0		2.0		255			1	Pedestrian Pushbutton / Cycle Detector 4
15		I	Y 2		1		2.0		255			1	Pedestrian Kerbside detector 4 (*)
16	PV1A	I			0				255			1	Local Link PV1 vehicle hold input (Phase A)
17		I			0				255			2	Local link PV1 vehicle hold input (Phase C)
18	DOOR	I			0				255			1+2	Cabinet Door Input Closed = Closed Circuit
19		I			0				255			1+2	Synchronise controller time to 12:00:00
20	AUD_ML1	I			0				255			1	Monitor Audible Drive - Loud (*)
21	AUD_MQ1	I			0				255			1	Monitor Audible Drive - Quiet (*)
22	AUD_ML2	I			0				255			2	Monitor Audible Drive - Load (*)
23	AUD_MQ2	I			0				255			2	Monitor Audible Drive - Quiet (*)
24		O			0				255			1	Link Out (Pedestrian Inhibit)
25		O			0				255			2	Link Out (Pedestrian Inhibit)
26		O			0				255			1	LLDEM1 Confirm (*)
27		O			0				255			1/2	LLDEM2 Confirm (*)

* See section 2.1.5 for Notes.

Column Titles:
N = The 'logical' I/O line number of the I/O function used as the index to identify the function by most I/O commands.
DFM = Detector Fault Monitoring settings - DFS+DFA ('Y','A','I' or blank if DFM disabled) and DGP (DFM Group 0-7).
IPS = Input sense, where '1' indicates I/O line is inverted.
Note: Kerbside and On-Crossing inputs are inverted by default to use 'open for detect' for safety.
IPX = Input extension time.
IOA = I/O allocation - the physical I/O line number or 255 if the I/O function is not used/not required.
Term. Block = Record in this column the terminal block position of the physical I/O line.
Pel = Shows whether the function applies to the first, the second or to both pelican streams.

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2.1.2) I/O Functions Provided (UTC)
=====

N	Mnemonic	In/Out	Def DFM	New DFM	Def IPS	New IPS	Def IPX	New IPX	Def IOA	New IOA	Term. Block	Pel	Description
32	LLDEM1	I			0				255			1	UTC PV vehicle hold input (Phase A)
33		I			0				255			1	UTC Pedestrian Demand (Phase B)
34		I			0				255			1	UTC switch to Fixed Vehicle Period Operation
35		I			0				255			1	UTC vehicle extension inhibit
36		I			0				255			1	UTC Local link disable
37		I			0				255			1	Local link demand (*)
38		I			0				255			1+2	Solar override (but see supply dimming page)
39		I			0				255			1+2	UTC take over bit
40		I			0				255			2	UTC PV vehicle hold input (Phase C)
41		I			0				255			2	UTC Pedestrian Demand (Phase D)
42		I			0				255			2	UTC switch to Fixed Vehicle Period Operation
43		I			0				255			2	UTC vehicle extension inhibit
44		I			0				255			2	UTC Local link disable
45		I			0				255			1/2	Local link demand (*)
46		I			0				255			1+2	UTC force max green timeset B
47		I			0				255			1+2	Fall back mode
48	For Inputs 48 to 55, see the next page												
56		O			1				255			1	Vehicle Green confirm (*)
57		O			1				255			1	Pedestrian Green confirm (*)
58		O			1				255			1	Wait Inidcator Active confirm (*)
59		O			1				255			1	UTC Ped Clearance Reply (*)
60		O			0				255			1	Local Link disable confirm
61		O			0				255			1+2	Detector Fault
62		O			0				255			1+2	Signals Bright Confirm
63		O			0				255			1+2	Remote Reconnect
64		O			1				255			2	Vehicle Green confirm (*)
65		O			1				255			2	Pedestrian Green confirm (*)
66		O			1				255			2	Wait Inidcator Active confirm (*)
67		O			1				255			2	UTC Ped Clearance Reply (*)
68		O			0				255			2	Local Link disable confirm
69		O			0				255			1+2	Controller Fault (*)
70		O			0				255			1+2	Clock 'RTCSYNC' confirm (set for 3 secs)
71		O			0				255			1+2	Fall Back Confirm

* See section 2.1.5 for Notes.

Column Titles:
N = The 'logical' I/O line number of the I/O function used as the index to identify the function by most I/O commands.
DFM = Detector Fault Monitoring settings - DFS+DFA ('Y','A','I' or blank if DFM disabled) and DGP (DFM Group 0-7).
IPS = Input sense, where '1' indicates I/O line is inverted (Also see note below on GXA etc.)
IPX = Input extension time.
IOA = I/O allocation - the physical I/O line number or 255 if the I/O function is not used/not required.
Term. Block = Record in this column the terminal block position of the physical I/O line.
Pel = Shows whether the function applies to the first, the second or to both pelican streams.

Special Instructions

2.1.3) I/O Functions Provided (Detection & SDE/SA)

N	Mnemonic	In/Out	DefDFM	NewDFM	DefIPS	NewIPS	DefIPX	NewIPX	DefIOA	NewIOA	Term. Block	Pel	Description
48		I	Y 0		0		0.0		255			2	Vehicle detector input 1
49		I	Y 0		0		0.0		255			2	Vehicle detector input 2
50		I	Y 3		1		0.6		255			2	Pedestrian On-Crossing Detector 1
51		I	Y 3		1		0.6		255			2	Pedestrian On-Crossing Detector 2
52		I	Y 1		0		2.0		255			2	Pedestrian Pushbutton / Cycle Detector 1
53		I	Y 2		1		2.0		255			2	Pedestrian Kerbside detector 1 (*)
54		I	Y 1		0		2.0		255			2	Pedestrian Pushbutton / Cycle Detector 2
55		I	Y 2		1		2.0		255			2	Pedestrian Kerbside detector 2 (*)
72		I	Y 0		0		0.0		255			2	Vehicle detector input 3
73		I	Y 0		0		0.0		255			2	Vehicle detector input 4
74		I	Y 3		1		0.6		255			2	Pedestrian On-Crossing Detector 3
75		I	Y 3		1		0.6		255			2	Pedestrian On-Crossing Detector 4
76		I	Y 1		0		2.0		255			2	Pedestrian Pushbutton / Cycle Detector 3
77		I	Y 2		1		2.0		255			2	Pedestrian Kerbside detector 3 (*)
78		I	Y 1		0		2.0		255			2	Pedestrian Pushbutton / Cycle Detector 4
79		I	Y 2		1		2.0		255			2	Pedestrian Kerbside detector 4 (*)
*** WHEN ALLOCATING SDE/SA INPUTS, BOTH LOOPS OF AN ASSESSOR MUST BE CONNECTED TO THE SAME I/O CARD OR BACKPLANE ***													
104		I	Y 0		0				255			1	SDE/SA Assessor 0 loop A (Phase A)
105		I	Y 0		0				255			1	SDE/SA Assessor 0 loop B (Phase A)
106		I	Y 0		0				255			1	SDE/SA Assessor 1 loop A (Phase A)
107		I	Y 0		0				255			1	SDE/SA Assessor 1 loop B (Phase A)
108		I	Y 0		0				255			1	SDE/SA Assessor 2 loop A (Phase A)
109		I	Y 0		0				255			1	SDE/SA Assessor 2 loop B (Phase A)
110		I	Y 0		0				255			1	SDE/SA Assessor 3 loop A (Phase A)
111		I	Y 0		0				255			1	SDE/SA Assessor 3 loop B (Phase A)
112		I	Y 0		0				255			2	SDE/SA Assessor 4 loop A (Phase C)
113		I	Y 0		0				255			2	SDE/SA Assessor 4 loop B (Phase C)
114		I	Y 0		0				255			2	SDE/SA Assessor 5 loop A (Phase C)
115		I	Y 0		0				255			2	SDE/SA Assessor 5 loop B (Phase C)
116		I	Y 0		0				255			2	SDE/SA Assessor 6 loop A (Phase C)
117		I	Y 0		0				255			2	SDE/SA Assessor 6 loop B (Phase C)
118		I	Y 0		0				255			2	SDE/SA Assessor 7 loop A (Phase C)
119		I	Y 0		0				255			2	SDE/SA Assessor 7 loop B (Phase C)

* See section 2.1.5 for Notes.

Column Titles:
N = The 'logical' I/O line number of the I/O function used as the index to identify the function by most I/O commands.
DFM = Detector Fault Monitoring settings - DFS+DFA ('Y','A','I' or blank if DFM disabled) and DGP (DFM Group 0-7).
IPS = Input sense, where '1' indicates I/O line is inverted.
Note: Kerbside and On-Crossing inputs are inverted by default to use 'open for detect' for safety.
IPX = Input extension time.
IOA = I/O allocation - the physical I/O line number or 255 if the I/O function is not used/not required.
Term. Block = Record in this column the terminal block position of the physical I/O line.
Pel = Shows whether the function applies to the first, the second or to both pelican streams.

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Special Instructions

2.1.4) I/O Functions Provided (Misc Outputs)
=====

N	Mnemonic	In/Out	Def DFM	New DFM	Def IPS	New IPS	Def IPX	New IPX	Def IOA	New IOA	Term. Block	Pel	Description
160		O			0				255			1+2	Any lamp failure (on either stream)
161		O			0				255			1	Any lamp failure
162		O			0				255			2	Any lamp failure
163		O			0				255			1	First vehicle red lamp failure
164		O			0				255			1	Second vehicle red lamp failure
165		O			0				255			2	First vehicle red lamp failure
166		O			0				255			2	Second vehicle red lamp failure
167	MATTEST	O			0				255			1+2	Kerbside Detector Mat Test Output (*)
168	LAMPSUP	O			0				255			1+2	Lamp Supply Pres & no RLM flt (either strm)
170		O			0				255			1+2	Lamps Extinguished inc RLM (either stream)
171		O			0				255			1	Lamps Extinguished (signals off or RLM flt)
172		O			0				255			2	Lamps Extinguished (signals off or RLM flt)

* See section 2.1.5 for Notes.

Column Titles:
N = The 'logical' I/O line number of the I/O function used as the index to identify the function by most I/O commands.
DFM = Detector Fault Monitoring settings - DFS+DFA ('Y','A','I' or blank if DFM disabled) and DGP (DFM Group 0-7).
IPS = Input sense, where '1' indicates I/O line is inverted.
IPX = Input extension time.
IOA = I/O allocation - the physical I/O line number or 255 if the I/O function is not used/not required.
Term. Block = Record in this column the terminal block position of the physical I/O line.
Pel = Shows whether the function applies to the first, the second or to both pelican streams.

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Special Instructions

2.1.5) I/O Functions Provided (Notes)

Pushbutton and Kerbside Detectors (8:PEL1PB1 & 9:PEL1KBS1, etc.)

If kerbside detectors are used, each pushbutton must be connected to its own dedicated input. Pushbuttons must not be connected in parallel to the same input. Each kerbside detector input is paired with one pushbutton input:
PEL1PB<x> is paired with PEL1KBS<x>
For example, the kerbside detector monitoring the area around the location of the pushbutton connected to PEL1PB1 must be connected to PEL1KBS1. If a pushbutton/cycle detector does not have a kerbside detector (e.g. cycle loop), leave its paired kerbside detector input de-allocated and the firmware will automatically insert a latched demand. If only one kerbside detector is monitoring the area around two pushbutton inputs (e.g. PEL1PB1 & PEL1PB2) assign the kerbside detector inputs (e.g. PEL1KBS1 & PEL1KBS2) for both pushbuttons to the same physical input to which the one kerbside detector is connected.

Monitor Audible Drives (20:AUD_ML1, 21:AUD_MQ1, 22:AUD_ML2, 23:AUD_MQ2)

See section 1.3

Local link Demand (37:LLDEM1, 45:LLDEM2, 26:BUSOUT1, 27:BUSOUT2)

By default, the inputs LLDEM1 and LLDEM2 insert latched Pedestrian demands.
If CFE1=1 is entered, these inputs will function as call/cancel pedestrian demands (see 7.2 - Ped Timings)
If CFE2=1 is entered, these inputs provide an alternative 'vehicle hold' function (see 6.1 - Vehicle Timings)
(BUSOUT1 & BUSOUT2 are the corresponding confirm outputs when used with CFE2:1)

Vehicle/Ped/Wait/Clearance Confirm (56:GXA,57:PCB,58:WIB,59:PRB, 64:GXC,65:PCD,66:WID,67:PRD)

Important: In order to meet TR2523A, if the above I/O lines are used they must be configured to normally closed contact outputs. Section "2.2) Provision of Physical I/O Lines" which follows the "I/O Functions Provided" tables shows the location of the controller's normally closed contact outputs.

Controller Fault (69:CF)

CF includes DFM faults unless the DF output is assigned to a physical output using IOA.
CF includes lamp faults unless LF or LF1 is assigned to a physical output using IOA.
If CFE3=1 is entered, CF goes active if the signals are off for any reason.

Kerbside Detector Mat Test Output (167:MATTEST)

If surface kerbside detectors are used, the 'MATTEST' output should be allocated to a normally open contact output, with one side of the output connected to the 24v detector supply and the other side connected to the test input of each mat detector. If above ground kerbside detectors are used, then the 'MATTEST' output should be left de-allocated.

Lamp Supply Output for use with a serially connected OMU (168:LAMPSUP)

This Output goes open circuit when there is no Lamp Supply or there is a Second Red Lamp failure.
It is used to indicate that the Lamp Supply is Present on an ELV controller which is connected serially to an OMU.
If the serial link to the OMU is lost, the LAMPSUP output indicates the Lamp Supply status to the OMU via an external relay. Assign the LAMPSUP output to an available I/O card output using IOA (IOA168=xx, where xx is the DET No. of the output used). Using the Kit 667/1/32612/000, connect the external relay to this output and to the OMU (in accordance with the instructions on drawing 667/GA/32612/000).

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2.1.6) I/O Overview

The table below summarises the I/O available on the logical I/O ports, showing the port number and range of detector numbers on that port, e.g. port '1' uses detector numbers '15' through to '8'.
When used with the IOL handset command, it shows the name of each of the eight I/O lines displayed on each port.
For example, IOL 0 displays the status of eight I/O lines, with the right-most bit showing the status of PEL1DET1.

Port	Det	I/O Lines on that port...								Det
IOL 0	7									0
IOL 1	15									8
IOL 2	23									16
IOL 3	31									24
IOL 4	39									32
IOL 5	47									40
IOL 6	55									48
IOL 7	63									56
IOL 8	71									64
IOL 9	79									72
IOL 13	111									104
IOL 14	119									112
IOL 20	167									160
IOL 21	175									168

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Special Instructions

2.2.2.1) Provision Of Physical I/O Lines (Built-In)

The following physical inputs are provided by the built-in ST750ELV PSU-I/O Card...

IOR	I/O	Type	IOA / Mnemonic	Provided By	Terminal (on ST750ELV Backplane)
0	I	input		SIO #1	0 "DIGITAL INPUT"
1	I	input		SIO #1	1
2	I	input		SIO #1	2
3	I	input		SIO #1	3
4	I	input		SIO #1	4
5	I	input		SIO #1	5
6	I	input		SIO #1	6
7	I	input		SIO #1	7 "DIGITAL INPUT"
8	I	input		SIO #1	8 "DIGITAL INPUT"
9	I	input		SIO #1	9
10	I	input		SIO #1	10
11	I	input		SIO #1	11
12	I	input		SIO #1	12
13	I	input		SIO #1	13
14	I	input		SIO #1	14
15	I	input		SIO #1	15 "DIGITAL INPUT"
16	I	input		SIO #1	16 "DIGITAL INPUT"
17	I	input		SIO #1	17
18	I	input		SIO #1	18
19	I	input		SIO #1	19
20	I	input		SIO #1	20
21	I	input		SIO #1	21
22	I	input		SIO #1	22
23	I	input		SIO #1	23 "DIGITAL INPUT"
24	O	output		SIO #1	8 NO1/COM/NC1 "RELAY"
25	O	output		SIO #1	9 NO2/COM/NC2
26	O	output		SIO #1	10 NO3/COM/NC3
27	O	output		SIO #1	11 NO4/COM/NC4 "RELAY"

Note 1: The 'IOR' column shows the 'physical I/O line number', i.e. the value of 'y' in the handset commands 'IOA x:y' and 'IOR y:x'. Use the handset command IOR to determine which I/O function (if any) has been allocated to each physical I/O line and enter these in to the IOA/Mnemonic column.

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Special Instructions

2.2.2) Provision Of Physical I/O Lines (Serial I/O Card 24/16)
=====

The following physical inputs are provided if a 24/16 Serial I/O Card is fitted.
NB: Only fit a 24/16 card if more than eight outputs are required (Four built-in plus a 24/4 gives eight outputs)
If this card is required, enter IOB 2=1 and select address #2 on the Serial I/O Card.

IOR	I/O	Type	IOA / Mnemonic	Provided By	Terminal (Typical; correct if necessary)
32	I	input		SIO #2	1I/02 0
33	I	input		SIO #2	1I/02 1
34	I	input		SIO #2	1I/02 2
35	I	input		SIO #2	1I/02 3
36	I	input		SIO #2	1I/02 4
37	I	input		SIO #2	1I/02 5
38	I	input		SIO #2	1I/02 6
39	I	input		SIO #2	1I/02 7
40	I	input		SIO #2	1I/02 8
41	I	input		SIO #2	1I/02 9
42	I	input		SIO #2	1I/02 10
43	I	input		SIO #2	1I/02 11
44	I	input		SIO #2	1I/02 12
45	I	input		SIO #2	1I/02 13
46	I	input		SIO #2	1I/02 14
47	I	input		SIO #2	1I/02 15
48	I	input		SIO #2	1I/02 16
49	I	input		SIO #2	1I/02 17
50	I	input		SIO #2	1I/02 18
51	I	input		SIO #2	1I/02 19
52	I	input		SIO #2	1I/02 20
53	I	input		SIO #2	1I/02 21
54	I	input		SIO #2	1I/02 22
55	I	input		SIO #2	1I/02 23
56	O	output		SIO #2	1I/02 0
57	O	output		SIO #2	1I/02 1
58	O	output		SIO #2	1I/02 2
59	O	output		SIO #2	1I/02 3
60	O	output		SIO #2	1I/02 4
61	O	output		SIO #2	1I/02 5
62	O	output		SIO #2	1I/02 6
63	O	output		SIO #2	1I/02 7
64	O	output		SIO #2	1I/02 8
65	O	output		SIO #2	1I/02 9
66	O	output		SIO #2	1I/02 10
67	O	output		SIO #2	1I/02 11
68	O	output		SIO #2	1I/02 12
69	O	output		SIO #2	1I/02 13
70	O	output		SIO #2	1I/02 14
71	O	output		SIO #2	1I/02 15

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Special Instructions

2.2.3) Provision Of Physical I/O Lines (Serial I/O Card 24/4)

The following physical inputs are provided if a 24/4 Serial I/O Card is fitted.
If this card is required, enter IOB 3=1 and select address #3 on the Serial I/O Card.

IOR	I/O	Type	IOA / Mnemonic	Provided By	Terminal (Typical; correct if necessary)
72	I	input		SIO #3	1I/02 0
73	I	input		SIO #3	1I/02 1
74	I	input		SIO #3	1I/02 2
75	I	input		SIO #3	1I/02 3
76	I	input		SIO #3	1I/02 4
77	I	input		SIO #3	1I/02 5
78	I	input		SIO #3	1I/02 6
79	I	input		SIO #3	1I/02 7
80	I	input		SIO #3	1I/02 8
81	I	input		SIO #3	1I/02 9
82	I	input		SIO #3	1I/02 10
83	I	input		SIO #3	1I/02 11
84	I	input		SIO #3	1I/02 12
85	I	input		SIO #3	1I/02 13
86	I	input		SIO #3	1I/02 14
87	I	input		SIO #3	1I/02 15
88	I	input		SIO #3	1I/02 16
89	I	input		SIO #3	1I/02 17
90	I	input		SIO #3	1I/02 18
91	I	input		SIO #3	1I/02 19
92	I	input		SIO #3	1I/02 20
93	I	input		SIO #3	1I/02 21
94	I	input		SIO #3	1I/02 22
95	I	input		SIO #3	1I/02 23
96	O	output		SIO #3	1I/02 0
97	O	output		SIO #3	1I/02 1
98	O	output		SIO #3	1I/02 2
99	O	output		SIO #3	1I/02 3

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Special Instructions

2.2.4) Provision Of Physical I/O Lines (Detector Backplanes)

The following physical inputs are provided if one Intelligent Detector Backplane is fitted
If this card is required, enter IOB 4=1 and select address #4 on the Intelligent Detector Backplane.

IOR	I/O	Type	IOA / Mnemonic	Provided By	Terminal (Typical; correct if necessary)
104	I	input		B/P #4	2LT1 A1
105	I	input		B/P #4	2LT1 A2
106	I	input		B/P #4	2LT1 A3
107	I	input		B/P #4	2LT1 A4
108	I	input		B/P #4	2LT1 B1
109	I	input		B/P #4	2LT1 B2
110	I	input		B/P #4	2LT1 B3
111	I	input		B/P #4	2LT1 B4
112	I	input		B/P #4	2LT1 C1
113	I	input		B/P #4	2LT1 C2
114	I	input		B/P #4	2LT1 C3
115	I	input		B/P #4	2LT1 C4
116	I	input		B/P #4	2LT1 D1
117	I	input		B/P #4	2LT1 D2
118	I	input		B/P #4	2LT1 D3
119	I	input		B/P #4	2LT1 D4

The following physical inputs are provided if a second Intelligent Detector Backplane is fitted.
If this card is required, enter IOB 5=1 and select address #5 on the Intelligent Detector Backplane.

IOR	I/O	Type	IOA / Mnemonic	Provided By	Terminal (Typical; correct if necessary)
120	I	input		B/P #5	2LT2 A1
121	I	input		B/P #5	2LT2 A2
122	I	input		B/P #5	2LT2 A3
123	I	input		B/P #5	2LT2 A4
124	I	input		B/P #5	2LT2 B1
125	I	input		B/P #5	2LT2 B2
126	I	input		B/P #5	2LT2 B3
127	I	input		B/P #5	2LT2 B4
128	I	input		B/P #5	2LT2 C1
129	I	input		B/P #5	2LT2 C2
130	I	input		B/P #5	2LT2 C3
131	I	input		B/P #5	2LT2 C4
132	I	input		B/P #5	2LT2 D1
133	I	input		B/P #5	2LT2 D2
134	I	input		B/P #5	2LT2 D3
135	I	input		B/P #5	2LT2 D4

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Special Instructions

2.2.5) Wiring for Intelligent Detector Backplanes
=====

The 24v DC detector supply for Loop Detectors and/or AGDs is derived from a winding on the Lamp Supply transformer. This voltage is rectified is output at DETECT+ and DETECT- with a maximum current capability of 2.0 amps. For details refer to the handbook 667/HB/33700/000.

If there is insufficient capability left on 24v DC to power the required number of loop detectors and/or AGDs the easiest option is to power them all from an ancillary 24v DC supply & mounting kit:
667/1/33074/000 24V DC 6A

(BACKPLANE 1)				(BACKPLANE 2)			
SERIAL CONNECTION				SERIAL CONNECTION			
RJ45 COMM'S CABLE				RJ45 COMM'S CABLE			
(NOTE1)	BACKPLANE			BACKPLANE			
CPU PCB	IN	OUT		IN	OUT		
PL2/4--- (RJ45) ----			(RJ45) -----				etc
POWER CONNECTION				POWER CONNECTION			
BACKPLANE				BACKPLANE			
POWER	IN+	OUT+		IN+	OUT+		
DETECT+ -- (RED) ---	SK7	SK6-----	(RED) -----	SK7	SK6		etc
IN- OUT-				IN- OUT-			
DETECT- -- (BLK) ---	SK7	SK6-----	(BLK) -----	SK7	SK6		etc

Note 1 For ST750ELV, one IO RJ45 connection from the CPU PCB is always connected to the ST750ELV backplane for the built-in I/O card (SIO#1).

This leaves one RJ45 connection on the CPU card that must daisy-chain to the optional I/O Cards and Detector Backplanes in any order convenient for the positions in the cabinet.

LOOP TERMINATION BOARD LT1 = DETECTOR BACKPLANE ADDRESS: #4			
LOOP DESIGNATION AND CONNECTIONS			
DET CARD	DET CARD	DET CARD	DET CARD
A	B	C	D
No. LOOP ID	No. LOOP ID	No. LOOP ID	No. LOOP ID
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
LOOP TERMINATION BOARD LT2 = DETECTOR BACKPLANE ADDRESS: #5			
LOOP DESIGNATION AND CONNECTIONS			
DET CARD	DET CARD	DET CARD	DET CARD
A	B	C	D
No. LOOP ID	No. LOOP ID	No. LOOP ID	No. LOOP ID
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4

Works Order :
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Special Instructions

2.2.6) Wiring for Single Detector Backplanes
=====

If single detector backplanes are required, rather than Intelligent Detector Backplanes, then use these pages.

SIEMENS SINGLE INTERNAL DETECTOR BACKPLANE INSTRUCTIONS SHEET (BACKPLANE:)

CONNECTIONS MADE USING CABLEFORM 667/1/03887/002
UNUSED WIRE ENDS MUST BE TIED BACK AND INSULATED

DETECTOR RACK POWER CONNECTIONS

SIGNAL	WIRE COLOUR	CONNECTION FROM CONTROLLER	BACKPLANE TERMINALS
24 VOLTS	RED	----	19
0 VOLTS	BLACK	----	20
SCREEN	PINK		22
COMMON	WHITE		18
24 v AC1	Orange/Grey		23
24 v AC2	Orange/Grey		24

Power linking between backplanes to be made using Orange & Grey wire from 667/1/03887/002

LOOP INPUTS

LOOP No.	DESIGNATION	BACKPLANE TERMINALS
1		1 & 2
2		3 & 4
3		5 & 6
4		7 & 8

DETECTOR OUTPUTS

DETECTOR No.	BACKPLANE TERMINALS	COLOUR	CONTR TERMINALS
1	10	BLUE	
2	12	GREEN	
3	14	ORANGE	
4	16	YELLOW	

[Template - Internal Detectors.txt Issue 2.0]

Works Order :
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Special Instructions

SIEMENS SINGLE INTERNAL DETECTOR BACKPLANE INSTRUCTIONS SHEET (BACKPLANE:)

CONNECTIONS MADE USING CABLEFORM 667/1/03887/002
UNUSED WIRE ENDS MUST BE TIED BACK AND INSULATED

DETECTOR RACK POWER CONNECTIONS			
SIGNAL	WIRE COLOUR	CONNECTION FROM CONTROLLER	BACKPLANE TERMINALS
24 VOLTS	RED	----	19
0 VOLTS	BLACK	----	20
SCREEN	PINK		22
COMMON	WHITE		18
24 v AC1	Orange/Grey		23
24 v AC2	Orange/Grey		24

LOOP INPUTS		
LOOP No.	DESIGNATION	BACKPLANE TERMINALS
1		1 & 2
2		3 & 4
3		5 & 6
4		7 & 8

DETECTOR OUTPUTS			
DETECTOR No.	BACKPLANE TERMINALS	COLOUR	CONTR TERMINALS
1	10	BLUE	
2	12	GREEN	
3	14	ORANGE	
4	16	YELLOW	

Power linking between backplanes to be made using Orange & Grey wire from 667/1/03887/002

[Template - Internal Detectors.txt Issue 2.0]

Works Order :
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Engineer :
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Special Instructions

3.1) Controller I/O Handset Commands
=====

Command	Default	Altered	Function	Access
IOB 1	1	---	IOC#1 - Built-in ST750ELV PSU-I/O Card (24/4)	2
IOB 2	0		IOC#2 - Optional Serial I/O Card (24/16) } Only one 24/16 *OR* 24/4 Serial	2
IOB 3	0		IOC#3 - Optional Serial I/O Card (24/4) } I/O Card can be fitted.	2
IOB 4	0		IOC#4 - Optional Intelligent Detector Backplane	2
IOB 5	0		IOC#5 - Optional Intelligent Detector Backplane	2
IOA x	y	---	I/O Allocation - See the I/O tables on the previous pages	3
IOR y	x	---	Review I/O allocation changes - See the I/O tables on the previous pages	R
IPS x	---	---	I/O line to be inverted - See the I/O tables on the previous pages	2
IPX x	---	---	Input extension time - See the I/O tables on the previous pages	3
DFS x	---	---	DFM Suppressed, i.e. disabled - See the I/O tables on the previous pages	2
DFA x	---	---	DFM Force Action - See the I/O tables on the previous pages	2
DGP x	---	---	DFM Group - See the I/O tables on the previous pages	2
CFE 0	---	---	Conditioning Facility Enable: See referenced pages for details, e.g. 7.1 (7.1) Enables vehicle detector checking	3
CFE 1	---	---	(7.2) LLDEM1/LLDEM2 demand ped phase with call/cancel	3
CFE 2	---	---	(6.1) LLDEM1/LLDEM2 hold the vehicle stages	3
CFE 3	0		(2.1) 'CF' includes signals off for any reason	3
CFE 4	---	---	(7.2) Dual crossing: automatic forward ped demand	3
CFE 5	---	---	(7.2) Dual crossing: automatic forward ped demand and extension inhibit	3
CFE 6	---	---	(7.2) Enable audible outputs and checking	3

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Special Instructions

3.2) Detector Fault Monitoring (Groups 0-3)

Command	Default	Altered	Function	Access
DSA 0 0	60		DFM Group 0 DFM timeset A fail active time in minutes (Vehicle Detectors)	2
DSA 0 1	60		DFM Group 0 DFM timeset B fail active time in minutes	2
DSA 0 2	60		DFM Group 0 DFM timeset C fail active time in minutes	2
DSA 0 3	60		DFM Group 0 DFM timeset D fail active time in minutes	2
DSI 0 0	18		DFM Group 0 DFM timeset A fail inactive time in hours	2
DSI 0 1	18		DFM Group 0 DFM timeset B fail inactive time in hours	2
DSI 0 2	18		DFM Group 0 DFM timeset C fail inactive time in hours	2
DSI 0 3	18		DFM Group 0 DFM timeset D fail inactive time in hours	2
DSA 1 0	5		DFM Group 1 DFM timeset A fail active time in minutes (Ped. Push-buttons)	2
DSA 1 1	5		DFM Group 1 DFM timeset B fail active time in minutes	2
DSA 1 2	5		DFM Group 1 DFM timeset C fail active time in minutes	2
DSA 1 3	5		DFM Group 1 DFM timeset D fail active time in minutes	2
DSI 1 0	240		DFM Group 1 DFM timeset A fail inactive time in hours	2
DSI 1 1	240		DFM Group 1 DFM timeset B fail inactive time in hours	2
DSI 1 2	240		DFM Group 1 DFM timeset C fail inactive time in hours	2
DSI 1 3	240		DFM Group 1 DFM timeset D fail inactive time in hours	2
DSA 2 0	90		DFM Group 2 DFM timeset A fail active time in minutes (Kerbside Detectors)	2
DSA 2 1	90		DFM Group 2 DFM timeset B fail active time in minutes	2
DSA 2 2	90		DFM Group 2 DFM timeset C fail active time in minutes	2
DSA 2 3	90		DFM Group 2 DFM timeset D fail active time in minutes	2
DSI 2 0	48		DFM Group 2 DFM timeset A fail inactive time in hours	2
DSI 2 1	48		DFM Group 2 DFM timeset B fail inactive time in hours	2
DSI 2 2	48		DFM Group 2 DFM timeset C fail inactive time in hours	2
DSI 2 3	48		DFM Group 2 DFM timeset D fail inactive time in hours	2
DSA 3 0	60		DFM Group 3 DFM timeset A fail active time in minutes (On-Crossing Dets)	2
DSA 3 1	60		DFM Group 3 DFM timeset B fail active time in minutes	2
DSA 3 2	60		DFM Group 3 DFM timeset C fail active time in minutes	2
DSA 3 3	60		DFM Group 3 DFM timeset D fail active time in minutes	2
DSI 3 0	18		DFM Group 3 DFM timeset A fail inactive time in hours	2
DSI 3 1	18		DFM Group 3 DFM timeset B fail inactive time in hours	2
DSI 3 2	18		DFM Group 3 DFM timeset C fail inactive time in hours	2
DSI 3 3	18		DFM Group 3 DFM timeset D fail inactive time in hours	2

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Special Instructions

3.3) Detector Fault Monitoring (Groups 4-7)

Command	Default	Altered	Function	Access
DSA 4 0	255		DFM Group 4 DFM timeset A fail active time in minutes (Spare)	2
DSA 4 1	255		DFM Group 4 DFM timeset B fail active time in minutes	2
DSA 4 2	255		DFM Group 4 DFM timeset C fail active time in minutes	2
DSA 4 3	255		DFM Group 4 DFM timeset D fail active time in minutes	2
DSI 4 0	255		DFM Group 4 DFM timeset A fail inactive time in hours	2
DSI 4 1	255		DFM Group 4 DFM timeset B fail inactive time in hours	2
DSI 4 2	255		DFM Group 4 DFM timeset C fail inactive time in hours	2
DSI 4 3	255		DFM Group 4 DFM timeset D fail inactive time in hours	2
DSA 5 0	255		DFM Group 5 DFM timeset A fail active time in minutes (Spare)	2
DSA 5 1	255		DFM Group 5 DFM timeset B fail active time in minutes	2
DSA 5 2	255		DFM Group 5 DFM timeset C fail active time in minutes	2
DSA 5 3	255		DFM Group 5 DFM timeset D fail active time in minutes	2
DSI 5 0	255		DFM Group 5 DFM timeset A fail inactive time in hours	2
DSI 5 1	255		DFM Group 5 DFM timeset B fail inactive time in hours	2
DSI 5 2	255		DFM Group 5 DFM timeset C fail inactive time in hours	2
DSI 5 3	255		DFM Group 5 DFM timeset D fail inactive time in hours	2
DSA 6 0	255		DFM Group 6 DFM timeset A fail active time in minutes (Spare)	2
DSA 6 1	255		DFM Group 6 DFM timeset B fail active time in minutes	2
DSA 6 2	255		DFM Group 6 DFM timeset C fail active time in minutes	2
DSA 6 3	255		DFM Group 6 DFM timeset D fail active time in minutes	2
DSI 6 0	255		DFM Group 6 DFM timeset A fail inactive time in hours	2
DSI 6 1	255		DFM Group 6 DFM timeset B fail inactive time in hours	2
DSI 6 2	255		DFM Group 6 DFM timeset C fail inactive time in hours	2
DSI 6 3	255		DFM Group 6 DFM timeset D fail inactive time in hours	2
DSA 7 0	255		DFM Group 7 DFM timeset A fail active time in minutes (Spare)	2
DSA 7 1	255		DFM Group 7 DFM timeset B fail active time in minutes	2
DSA 7 2	255		DFM Group 7 DFM timeset C fail active time in minutes	2
DSA 7 3	255		DFM Group 7 DFM timeset D fail active time in minutes	2
DSI 7 0	255		DFM Group 7 DFM timeset A fail inactive time in hours	2
DSI 7 1	255		DFM Group 7 DFM timeset B fail inactive time in hours	2
DSI 7 2	255		DFM Group 7 DFM timeset C fail inactive time in hours	2
DSI 7 3	255		DFM Group 7 DFM timeset D fail inactive time in hours	2

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Special Instructions

4.1) Lamp Monitoring Arrangement

The configuration uses the following lamp monitor sensors. All sensors are built-in; no torroids required.

- The table shows which phase (A-D) and which LSLS Outputs (Terminals) are monitored by each sensor.
- The default Load Type is shown. This can be changed using the 'KLT' handset command (see section 4.3)
- To set the number of Red Lamp Monitor channels, see section 4.2.

Sensor	Terminal	Description	Load Type (Default)	Load Type (New) - See 4.3
1 A	1 RED1	Vehicle Phase		
1 A	6 AMB	AND	1:STC Helios ELV	
1 A	7 GRN	RLM Channel 1		
2 A	2 RED2	RLM Channel 2	1:STC Helios ELV	
3 A	3 RED3	RLM Channel 3	1:STC Helios ELV	
4 A	4 RED4	RLM Channel 4	1:STC Helios ELV	
5 A	5 RED5	RLM Channel 5	1:STC Helios ELV	
6 B	8 REDM	Ped Red/Grn	3:STC Near-Side Indicator	
6 B	12 GRNM			
7 B	9 REDM	Additional	3:STC Near-Side Indicator	
7 B	13 GRNM	Ped Red/Grn		
8 B	10 DMND	Wait/Demand	2:STC Demand Indicator	
9 B	11 DMND	(Additional)	2:STC Demand Indicator	
-- B	14 TACT	Not	-	
-- B	15 TACT	Lamp	-	
-- B	16 ADRV	Monitored	-	
11 C	17 RED1	Vehicle Phase		
11 C	22 AMB	AND	1:STC Helios ELV	
11 C	23 GRN	RLM Channel 1		
12 C	18 RED2	RLM Channel 2	1:STC Helios ELV	
13 C	19 RED3	RLM Channel 3	1:STC Helios ELV	
14 C	20 RED4	RLM Channel 4	1:STC Helios ELV	
15 C	21 RED5	RLM Channel 5	1:STC Helios ELV	
16 D	24 REDM	Ped Red/Grn	3:STC Near-Side Indicator	
16 D	28 GRNM			
17 D	25 REDM	Additional	3:STC Near-Side Indicator	
17 D	29 GRNM	Ped Red/Grn		
18 D	26 DMND	Wait/Demand	2:STC Demand Indicator	
19 D	27 DMND	(Additional)	2:STC Demand Indicator	
-- D	30 TACT	Not	-	
-- D	31 TACT	Lamp	-	
-- D	32 ADRV	Monitored	-	

Care must be taken to not overload any one output by connecting too many signals. To allow more signals to be connected, two outputs are provided for each of the pedestrian aspects.

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Special Instructions

4.2) Red Lamp Monitoring

Five RLM channels for each crossing are enabled by default. If less than five RLM channels are required, use the handset command "RLM" (below) to reduce the number of channels.

The function of the sensors depends on how many RLM channels are configured. Sensors shown with a '-' when the number of channels are reduced do not perform red lamp monitoring. However, the sensor still performs lamp monitoring of the vehicle red signals connected and so can be used to detect lamp failures on 'unmonitored' vehicle red signals for reporting to UTC or RMS for example.

Terminals	RLM n:5	RLM n:4	RLM n:3	RLM n:2	RLM n:1	Lamp Monitor Sensor Location
RED1	Channel 1	Channel 1	Channel 1	Channel 1	Channel 1	Built in to the driver card
RED2	Channel 2	Channel 2	Channel 2	Channel 2	-	Built in to the driver card
RED3	Channel 3	Channel 3	Channel 3	-	-	Built in to the driver card
RED4	Channel 4	Channel 4	-	-	-	Built in to the driver card
RED5	Channel 5	-	-	-	-	Built in to the driver card

For the location of the Terminals RED1,RED2,... refer to section 4.1

4.3) Lamp Monitoring Configuration Commands

Command	Default	Altered	Function	Access
RLM 0	5		Number of RLM channels (approaches) on first stream	3
RLM 1	5		Number of RLM channels (approaches) on second stream	3
			Enter "RLM 1=0" to extinguish the second stream if only a single is required.	
KLT s	---	---	Load Type for sensor 's'. See table in section 4.1	3
			Values: 0 = Disable Monitoring (e.g. if signal type cannot be monitored)	
			1 = STC Helios ELV (e.g. Vehicle and Far-Side Ped Signals)	
			2 = STC Demand Indicator (e.g. Puffin Demand Accepted Signals)	
			3 = STC Near-Side Indicator (e.g. Near-Side Puffin Signals)	
			4 = STC ELV Reg Sign	
			5 = STC LED Wait (e.g. Used with far-side ped signals)	
			Other types may be added at a later date; check handbook.	

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Special Instructions

5.1) Speed Discrimination / Speed Assessment

This configuration provides for up to 4 assessors (8 loops with 12 foot spacing) of speed discrimination on each stream. By default however, none of the SDE/SA detector inputs are allocated to physical I/O lines causing the fault "FLF 51:255 SDEN" to be raised.

If SDE/SA is not required, then enter 'SDS=0' and power the controller off and back on.

If SA is required, then change the value of SDS to '1'. If SDE is required, then ensure SDS is set to '2' (which is its default value) and configure the 'SDE Assessor Types' using the SAT handset command. After the value, the SAT command will display the associated phase and the assessor type as text, e.g. "SAT 0:2 (A,TInn)".

Each required SDE or SA detector input must then be assigned to a physical I/O line using the IOA command and the I/O tables on previous pages updated. If not all of the speed detector inputs are required, the unused ones should be left de-allocated.

*** BOTH INPUTS OF AN ASSESSOR MUST BE CONNECTED TO THE SAME I/O CARD OR BACKPLANE ***
It is recommended that the two inputs of an assessor use adjacent physical inputs.

Note: If any SDE/SA data is changed, the fault flag "FLF 52:255 SDED" will be set and artificial SDE/SA extensions are inserted until the controller's power is switched off and back on and the controller begins to use the new data.

Command	Default	Altered	Function	Access
SDS	2		SDE/SA Select: 0=None, 1=SA, 2=SDE	3
SDD 0	0		SDE/SA Disabled on Stream 0 (Use 'SDD 0=1' if SDE/SA only req. on stream 1)	3
SDD 1	0		SDE/SA Disabled on Stream 1 (Use 'SDD 1=1' if SDE/SA only req. on stream 0)	3
SAT 0	2		SDE Assessor Type: 1=Double, 2=Triple (inner), 3=Triple (outer) [Phase A]	3
SAT 1	3		SDE Assessor Type: 1=Double, 2=Triple (inner), 3=Triple (outer) [Phase A]	3
SAT 2	2		SDE Assessor Type: 1=Double, 2=Triple (inner), 3=Triple (outer) [Phase A]	3
SAT 3	3		SDE Assessor Type: 1=Double, 2=Triple (inner), 3=Triple (outer) [Phase A]	3
SAT 4	2		SDE Assessor Type: 1=Double, 2=Triple (inner), 3=Triple (outer) [Phase C]	3
SAT 5	3		SDE Assessor Type: 1=Double, 2=Triple (inner), 3=Triple (outer) [Phase C]	3
SAT 6	2		SDE Assessor Type: 1=Double, 2=Triple (inner), 3=Triple (outer) [Phase C]	3
SAT 7	3		SDE Assessor Type: 1=Double, 2=Triple (inner), 3=Triple (outer) [Phase C]	3

- Use the handset command 'SPE' to commission each SDE or SA assessor pair of inputs:
- Use IOA and check that required assessor inputs have been allocated to physical inputs.
 - Use SPE and check that the correct loops have been connected to the correct physical inputs.
 - Use SPE and check that the 'a' loop is the first loop that the vehicles cross.
 - Use SPE and check that the handset display matches the speed of the vehicle.
 - Use SAT and check that the associated phase and assessor type are correct.

Special Instructions

6.1) Vehicle Timings

=====

NB: Phase A is the vehicle phase in the first pedestrian stream and Phase C is the vehicle in the second

Command	Default	Altered	Function	Access
PEV 0	1		Ped Stream, Enable VA mode	3
PEV 1	1		PEV n=0 - Disables VA mode so the stream only ever runs FVP mode PEV n=1 - Enables VA mode, but manual panel (if fitted) may select either PEV n=2 - Forces VA mode (ignoring the manual panel) Important: Note that by default, this configuration will run VA mode. If no vehicle detectors are fitted and the controller should only run FVP mode, then "PEV n=0" must be entered.	
PTM A	0		Pre-Timed Maximum Enable \ PTM p=1 enables pre-timed maximum green, i.e. max	2
PTM C	0		Pre-Timed Maximum Enable / starts with vehicle green even if no ped demand.	2
PTX A	0		Pre-Timed Maximum Extra Period \ If ped demanded after pre-timed max expires	2
PTX C	0		Pre-Timed Maximum Extra Period / then max timer is restarted with this time. (Also see Pedestrian Timings - PDD)	2
MIN A	7		Vehicle Minimum Green Period	3
MIN C	7		Vehicle Minimum Green Period	3
EXT A	1.6		Vehicle Phase Green Extension Time	2
EXT C	1.6		Vehicle Phase Green Extension Time (Set to 0.0 if IPX is used to give each detector input its own extension time. Record each vehicle detector IPX time in the I/O tables)	2
MAX A	30		Vehicle Maximum Green Period (during timeset A)	2
MBX A	30		Vehicle Maximum Green Period (during timeset B)	2
MCX A	30		Vehicle Maximum Green Period (during timeset C)	2
MDX A	30		Vehicle Maximum Green Period (during timeset D)	2
MEX A	20		Fixed Vehicle Period (during timeset A)	2
MFX A	20		Fixed Vehicle Period (during timeset B)	2
MGX A	20		Fixed Vehicle Period (during timeset C)	2
MHX A	20		Fixed Vehicle Period (during timeset D)	2
MAX C	30		Vehicle Maximum Green Period (during timeset A)	2
MBX C	30		Vehicle Maximum Green Period (during timeset B)	2
MCX C	30		Vehicle Maximum Green Period (during timeset C)	2
MDX C	30		Vehicle Maximum Green Period (during timeset D)	2
MEX C	20		Fixed Vehicle Period (during timeset A)	2
MFX C	20		Fixed Vehicle Period (during timeset B)	2
MGX C	20		Fixed Vehicle Period (during timeset C)	2
MHX C	20		Fixed Vehicle Period (during timeset D)	2
CFE 2	0		Conditioning Facility Enable - CFE 2=1 causes LLDEM1/LLDEM2 to hold the vehicle stages (timings below) and not demand the ped phase.	3
CDT 1	5		Hold Phase A Vehicle Period when LLDEM1 goes active	2
CDT 2	0		Hold Phase A Vehicle Period when LLDEM2 goes active (For use on a single ped)	2
CDT 3	0		Hold Phase C Vehicle Period when LLDEM2 goes active (For use on a dual ped)	2

=====

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Special Instructions

7.1) Pedestrian Timings

NB: Phase B is the pedestrian phase in the first pedestrian stream and Phase D is the pedestrian in the second

Command	Default	Altered	Function	Access
PAR 0 0	1		Ped All-Red Period (Gap - vehicle phase gap-changed while running VA mode)	3
PAR 0 1	3		Ped All-Red Period (Max - vehicle phase ran to its VA maximum)	3
PAR 0 2	3		Ped All-Red Period (FVP - vehicle phase is running FVP mode)	3
PAR 0 3	3		Ped All-Red Period (UTC - stream under UTC 'PV' control)	3
PAR 0 4	3		Ped All-Red Period (Lnk - stream under CLF or Local Link 'PV1' control)	3
PAR 1 0	1		Ped All-Red Period (Gap - vehicle phase gap-changed while running VA mode)	3
PAR 1 1	3		Ped All-Red Period (Max - vehicle phase ran to its VA maximum)	3
PAR 1 2	3		Ped All-Red Period (FVP - vehicle phase is running FVP mode)	3
PAR 1 3	3		Ped All-Red Period (UTC - stream under UTC 'PV' control)	3
PAR 1 4	3		Ped All-Red Period (Lnk - stream under CLF or Local Link 'PV1' control)	3

			* Note: The pedestrian all-red period will always run to three seconds *	
			* if SDE/SA is configured. See section 5) in these instructions. *	

MIN B	5		Pedestrian Green Period	3
MIN D	5		Pedestrian Green Period	3
PBT B	3		Minimum Pedestrian Clearance Period	3
CMX B	10		Extendable Pedestrian Clearance Period	3
CDY B 0	1		Clearance Delay Period (after on-crossing detectors go inactive)	3
CDY B 1	3		Clearance Delay Period (if clearance period ran to its maximum)	3
PBT D	3		Minimum Pedestrian Clearance Period	3
CMX D	10		Extendable Pedestrian Clearance Period	3
CDY D 0	1		Clearance Delay Period (after on-crossing detectors go inactive)	3
CDY D 1	3		Clearance Delay Period (if clearance period ran to its maximum)	3
IPX det	0.6	---	Use IPX to change the extension time for each on-crossing detector input. Record new values in the I/O tables in section 2.1) I/O Functions Provided.	3

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Special Instructions

7.2) Advanced Pedestrian Timings

NB: Phase B is the pedestrian phase in the first pedestrian stream and Phase D is the pedestrian in the second

Command	Default	Altered	Function	Access
CIL	0		Enter 'CIL=1' to enable Cross Inhibit Linking.	2
IPX det	2.0	---	Use IPX to change the extension time for each pushbutton and kerbside. Record new values in the I/O tables in section 2.1) I/O Functions Provided.	3
PDX B	3.0		Demand timeout period - First Stream	2
PDX B	3.0		Demand timeout period - Second Stream	2
PDD B	1		Pedestrian Demand Delay - First Stream	2
PDD D	1		Pedestrian Demand Delay - Second Stream	2
CFE 0	0		Conditioning Facility Enable - Enter CFE 0 = 1 to enable special detector checking, a facility which if enabled, changes the pedestrian all-red period (PAR) from 'gap change' to 'max change' if any of the vehicle detectors remain inactive during the proceeding vehicle green period.	3
CFE 1	0		Conditioning Facility Enable - CFE 1=1 causes LLDEM1/LLDEM2 to use the call cancel times below. By default (CFE 1:0), these inputs insert latched demands for the associated ped phase.	3
DCL 0	3		LLDEM1 Call Delay \ First Stream	2
DCN 0	3		LLDEM1 Cancel Delay /	2
DCL 1	3		LLDEM2 Call Delay \ Second Stream	2
DCN 1	3		LLDEM2 Cancel Delay /	2
CFE 4	0		Conditioning Facility Enable - CFE 4=1 causes pushbutton demands on one crossing to automatically trigger a ped demand on the other crossing of this dual pedestrian crossing; also known as 'forward ped demand'. - PEL1PB1, PEL1PB2, PEL2PB1, PEL2PB2 are the 'outer' pushbuttons that will also insert demands for the other crossing if this facility is enabled. - PEL1PB3, PEL1PB4, PEL2PB3, PEL2PB4 are the 'inner' pushbuttons on the centre island that only demand their specific crossing.	3
CFE 5	0		Conditioning Facility Enable - CFE 5=1 causes vehicle extensions to be curtailed in addition to the ped demand triggered by CFE4 above.	3
CDT 10	10		PEL1 Time Set A \ Delay between start of green man on PEL1 to the insertion	2
CDT 11	10		PEL1 Time Set B \ of the automatic ped demand on PEL2, if PEL1PB1 or	2
CDT 12	10		PEL1 Time Set C / PEL1PB2 were pressed. See CFE4 and CFE5 above.	2
CDT 13	10		PEL1 Time Set D / One time value per max green timeset.	2
CDT 20	10		PEL2 Time Set A \ Delay between start of green man on PEL2 to the insertion	2
CDT 21	10		PEL2 Time Set B \ of the automatic ped demand on PEL1, if PEL2PB1 or	2
CDT 22	10		PEL2 Time Set C / PEL2PB2 were pressed. See CFE4 and CFE5 above.	2
CDT 23	10		PEL2 Time Set D / One time value per max green timeset.	2
CFE 6	0		Conditioning Facility Enable - CFE 6=1 enables monitoring of the audible	3
CDT 5	1.0		outputs using the inputs AUD_ML1/2 and AUD_MQ1/2. 'CDT 5' and 'CDT 6' set	
CDT 6	1.0		the confirmation time for PED1 and PED2 respectively. (Also see section 1.3)	

Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Special Instructions

8.1) UTC / Local Link / Dimming / Manual Panel
=====

Command	Default	Altered	Function	Access
UIE 0	1		UTC Inhibit Extensions Time (i.e. UTC 'PV' window time)	2
UIE 1	1		UTC Inhibit Extensions Time (i.e. UTC 'PV' window time)	2
RMR	0	---	RMR=1 set the 'RR' Remote Reconnect UTC reply bit Remember to enter 'RMR=0' to reset the RR bit when you leave site	2
LKD 0	0		PV1 Local Link Input (Stream 0) Delay time (seconds)	2
LKW 0	1		PV1 Local Link Input (Stream 0) Window time (seconds)	2
LKO 0	100		PV1 Local Link Input (Stream 0) Override time (seconds)	2
LKD 1	0		PV1 Local Link Input (Stream 1) Delay time (seconds)	2
LKW 1	1		PV1 Local Link Input (Stream 1) Window time (seconds)	2
LKO 1	100		PV1 Local Link Input (Stream 1) Override time (seconds)	2
LKM 0	0		Local Link Override Timer Mode (1=Enables self-resetting mode)	2
LKM 1	0		Local Link Override Timer Mode (1=Timer only starts when there is ped demand)	2
LKA 0	0		Local Link Fail Active Time (minutes) }	2
LKI 0	0		Local Link Fail Inactive Time (minutes) } Provide monitoring of the PV1	2
LKA 1	0		Local Link Fail Active Time (minutes) } input as described in MCE0145	2
LKI 1	0		Local Link Fail Inactive Time (minutes) }	2
LLD 0	0		Local Link Disable (1 disables the local link facility on stream 0)	2
LLD 1	0		Local Link Disable (1 disables the local link facility on stream 1)	2
KDP	1		Dimming Present (0=Disabled, 1=Enabled & Monitored, 2=Enabled but not mon.)	2
DIT	0		Dim Override Type for the 'SO' input (0=UTC force bright, 1=L/L force dim)	2
DOV	0		Dim Override (DOV:1 forces the signals to bright)	2
PEV	1	---	Ped Stream, Enable VA mode (see Vehicle Timings page)	3
MPA 0	18		Dim override button (force signals to bright)	3
MPA 1	255		Computer Control indicator (illuminated if PV active)	3
MPA 2	255		Disable Local Link button (inhibits PV1 operation)	3
MPA 3	19		Remote Reconnect button (activates the UTC RR reply bit)	3
			Use the following values to assign these functions to buttons/indicators:	
			MPA N:255 - Function 'N' not allocated to any buttons or indicators	
			MPA N:18 - SW1 button and AUX1 indicator provide function 'N'	
			MPA N:19 - SW2 button and AUX2 indicator provide function 'N'	
			MPA N:5 - SW3 button and AUX3 indicator provide function 'N'	
LPT	0		Manual panel (incl. cabinet alarm) lamp test	2

Special Instructions

9.1) Cableless Link Facility (CLF)
=====

This configuration has base time CLF enabled (base time is 2 AM each day) although no plans have been defined. Use this page to record any CLF plans that are entered using the handset:

Enter the plan number below, starting at zero. Use additional copies of this page for other plans.

Plan Number : ____ i.e. 'n' in the following table.

Command	Default	Altered	Function	Access
CYC n	0		CLF plan 'n' cycle time	2
OFF n 0	0		CLF plan 'n' offset time (seconds)	2
OFF n 1	0		CLF plan 'n' offset time (minutes)	2
PLI n	0		CLF plan 'n' runs which influence set (all plans will normally use set 0)	2
PLT n 0	255		CLF Plan 'n' group 0 - starts at this time in the cycle	2
IFN 0 0	9		Influence Set 0 group 0 - allows the pedestrian phase	2
IFS 0 0	2		Influence Set 0 group 0 - affects stage 2 (i.e. the first stream)	2
PLT n 1	255		CLF Plan 'n' group 1 - starts at this time in the cycle	2
IFN 0 1	8		Influence Set 0 group 1 - inhibits the pedestrian phase	2
IFS 0 1	2		Influence Set 0 group 1 - affects stage 2 (i.e. the first stream)	2
PLT n 2	255		CLF Plan 'n' group 2 - starts at this time in the cycle	2
IFN 0 2	9		Influence Set 0 group 2 - allows the pedestrian phase	2
IFS 0 2	4		Influence Set 0 group 2 - affects stage 4 (i.e. the second stream)	2
PLT n 3	255		CLF Plan 'n' group 3 - starts at this time in the cycle	2
IFN 0 3	8		Influence Set 0 group 3 - inhibits the pedestrian phase	2
IFS 0 3	4		Influence Set 0 group 3 - affects stage 4 (i.e. the second stream)	2
PLT n 4	255		CLF Plan 'n' group 4 -	2
IFN 0 4	255		Influence Set 0 group 4 -	2
IFS 0 4	255		Influence Set 0 group 4 -	2
PLT n 5	255		CLF Plan 'n' group 5 -	2
IFN 0 5	255		Influence Set 0 group 5 -	2
IFS 0 5	255		Influence Set 0 group 5 -	2
PLT n 6	255		CLF Plan 'n' group 6 -	2
IFN 0 6	255		Influence Set 0 group 6 -	2
IFS 0 6	255		Influence Set 0 group 6 -	2
PLT n 7	255		CLF Plan 'n' group 7 -	2
IFN 0 7	255		Influence Set 0 group 7 -	2
IFS 0 7	255		Influence Set 0 group 7 -	2

Intersection : Lower Bristol Road / Oak Street Puffin

Special Instructions

10.1) Controller Timetable

The controller's timetable is initially empty. Use this page to record any timetable entries that are entered using the handset command TSW:

TSW <Entry 0 to 63> <Index 0 to 5> : <Value 0 to 255>

This configuration uses the following day codes and operation codes:

TSW n 0:0	- Saturday	TSW n 4:0	TSW n 5:0	Isolate
TSW n 0:1	- Sunday	TSW n 4:1	TSW n 5:p	Run CLF plan 'p'
TSW n 0:2	- Monday			
TSW n 0:3	- Tuesday	TSW n 4:4	TSW n 5:0	Introduce maximum green timeset A
TSW n 0:4	- Wednesday	TSW n 4:3	TSW n 5:0	Introduce maximum green timeset B
TSW n 0:5	- Thursday	TSW n 4:3	TSW n 5:1	Introduce maximum green timeset C
TSW n 0:6	- Friday	TSW n 4:3	TSW n 5:2	Introduce maximum green timeset D
TSW n 0:7	- Everyday			
TSW n 0:8	- Everyday except Sunday	TSW n 4:4	TSW n 5:3	Introduce DFM timeset A
TSW n 0:9	- Monday to Friday	TSW n 4:3	TSW n 5:3	Introduce DFM timeset B
TSW n 0:10	- Not Used	TSW n 4:3	TSW n 5:4	Introduce DFM timeset C
TSW n 0:11	- Not Used	TSW n 4:3	TSW n 5:5	Introduce DFM timeset D
TSW n 0:12	- Not Used			
TSW n 0:13	- Not Used	TSW n 4:3	TSW n 5:6	Switch audibles off (or quiet)
TSW n 0:14	- Not Used	TSW n 4:4	TSW n 5:6	Switch audibles back on (or loud)
TSW n 0:15	- Not Used			

[illegible]

Works Order : XXXXXXXXXX
EM Number : 1084
Engineer : XXXXXXXXXX
Intersection : Lower Bristol Road / Oak Street Puffin

Special Instructions

Only two kerbside inputs are fitted on site, inputs BKSB3 and BKSB4 are currently unused. If required at a future date the inputs can be enabled using the IOA handset command:

BKBS3	IOA6=6	Default=255
BKSB4	IOA7=7	Default=255

Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Inputs and Outputs

Inputs and Outputs

☐ Enable Signal Required
Check boxes

☒ Manual Allocation

Port Number & Type

Port:

0

☐ Inputs

☐ Outputs

☒ Inputs & Outputs

Card Type & Address

Serial IO 24/4
Card Address 1

	DET No	Bit No	Type I or O	Name	Req'd	BP	Count	Inv	U/D	Misc	DFM	DFM Group	Ext time	Phs	UTC	SDE	Pri	Used By HC	CC	IG	UD	LRT	Term Block	Line No
☐	0	0	I	BPBU1	☒	☐	☐	☐	☐	☐	Y	2	0.0	☒	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-0
☐	1	1	I	BPBU2	☒	☐	☐	☐	☐	☐	Y	2	0.0	☒	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-1
☐	2	2	I	BPBU3	☒	☐	☐	☐	☐	☐	Y	2	0.0	☒	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-2
☐	3	3	I	BPBU4	☒	☐	☐	☐	☐	☐	Y	2	0.0	☒	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-3
☐	4	4	I	BKBS1	☒	☐	☐	☒	☐	☐	A	3	1.0	☒	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-4
☐	5	5	I	BKBS2	☒	☐	☐	☒	☐	☐	A	3	1.0	☒	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-5
☐	6	6	I	BKBS3	☐	☐	☐	☒	☐	☐	A	3	1.0	☒	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-6
☐	7	7	I	BKBS4	☐	☐	☐	☒		☐	A	3	1.0	☒	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-7

AddDeleteMoveClear Used ByMove to/from backplane

S.I. = See Special Instructions

Manual Map Optimisation

Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Inputs and Outputs

Inputs and Outputs

☐ Enable Signal Required
Check boxes

☒ Manual Allocation

Port Number & Type

Port:

1

☐ Inputs

☒ Inputs & Outputs

☐ Outputs

Card Type & Address

Serial IO 24/4
Card Address 1

	DET No	Bit No	Type I or O	Name	Req'd	BP	Count	Inv	U/D	Misc	DFM	DFM Group	Ext time	Phs	UTC	SDE	Used By				Pri	HC	CC	IG	UD	LRT	Term Block	Line No
☐	8	0	I	BONC1	☒	☐	☐	☒	☐	☐	A	4	1.0	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-8
☐	9	1	I	BONC3	☒	☐	☐	☒	☐	☐	A	4	1.0	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-9
☐	10	2	I	BTAC2	☒	☐	☐	☐	☐	☐	I	5	0.0	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-10
☐	11	3	I	BTAC4	☒	☐	☐	☐	☐	☐	I	5	0.0	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-11
☐	12	4	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-12
☐	13	5	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-13
☐	14	6	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-14
☐	15	7	I		☐	☐	☐	☐		☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-15

AddDeleteMoveClear Used ByMove to/from backplane

S.I. = See Special Instructions

Manual Map Optimisation

Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Inputs and Outputs

Inputs and Outputs

☐ Enable Signal Required
Check boxes

☒ Manual Allocation

Port Number & Type

Port:

2

☐ Inputs

☐ Outputs

☒ Inputs & Outputs

Card Type & Address

Serial IO 24/4
Card Address 1

	DET No	Bit No	Type I or O	Name	Req'd	BP	Count	Inv	U/D	Misc	DFM	DFM Group	Ext time	Phs	UTC	SDE	Pri	Used By HC	CC	IG	UD	LRT	Term Block	Line No
☐	16	0	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-16
☐	17	1	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-17
☐	18	2	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-18
☐	19	3	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-19
☐	20	4	I	AUD_ML1	☐	☐	☐	☐	☐	☒	N		0.0	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-20
☐	21	5	I	AUD_MQ1	☐	☐	☐	☐	☐	☒	N		0.0	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-21
☐	22	6	I	AUD_ML2	☐	☐	☐	☐	☐	☒	N		0.0	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-22
☐	23	7	I	AUD_MQ2	☐	☐	☐	☐		☒	N		0.0	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	I-23

AddDeleteMoveClear Used ByMove to/from backplane

S.I. = See Special Instructions

Manual Map Optimisation

Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Inputs and Outputs

Inputs and Outputs

☐ Enable Signal Required
Check boxes

☒ Manual Allocation

Port Number & Type

Port:

13

☐ Inputs

☐ Outputs

☒ Inputs & Outputs

Card Type & Address

Intelligent Backplane 160
Card Address 4

	DET No	Bit No	Type I or O	Name	Req'd	BP	Count	Inv	U/D	Misc	DFM	DFM Group	Ext time	Phs	UTC	SDE	Pri	Used By HC	CC	IG	UD	LRT	Term Block	Terminal No
☐	104	0	I	AIN1	☒	☒	☐	☐	☐	☐	A	0	32	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	105	1	I	AX2	☒	☒	☐	☐	☐	☐	A	0	36	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	106	2	I	AIN3	☒	☒	☐	☐	☐	☐	A	0	32	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	107	3	I	AX4	☒	☒	☐	☐	☐	☐	A	0	36	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	108	4	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	109	5	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	110	6	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	111	7	I		☐	☐	☐	☐		☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.

AddDeleteMoveClear Used ByMove to/from backplane

S.I. = See Special Instructions

Manual Map Optimisation

Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Inputs and Outputs

Inputs and Outputs

☐ Enable Signal Required
Check boxes

☒ Manual Allocation

Port Number & Type

Port:

15

☐ Inputs

☐ Outputs

☒ Inputs & Outputs

Card Type & Address

Intelligent Backplane 160
Card Address 5

	DET No	Bit No	Type I or O	Name	Req'd	BP	Count	Inv	U/D	Misc	DFM	DFM Group	Ext time	Phs	UTC	SDE	Pri	Used By HC	CC	IG	UD	LRT	Term Block	Terminal No
☐	120	0	I	DUMLOOP3	☐	☒	☐	☐	☐	☐	N		00	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	121	1	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	122	2	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	123	3	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	124	4	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	125	5	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	126	6	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	127	7	I		☐	☐	☐	☐		☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.

AddDeleteMoveClear Used ByMove to/from backplane

S.I. = See Special Instructions

Manual Map Optimisation

Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Inputs and Outputs

Inputs and Outputs

☐ Enable Signal Required Check boxes

☒ Manual Allocation

Port Number & Type

Port:

16

☐ Inputs

☐ Outputs

☒ Inputs & Outputs

Card Type & Address

Intelligent Backplane 160
Card Address 5

	DET No	Bit No	Type I or O	Name	Req'd	BP	Count	Inv	U/D	Misc	DFM	DFM Group	Ext time	Phs	UTC	SDE	Pri	Used By HC	CC	IG	UD	LRT	Term Block	Terminal No
☐	128	0	I	DUMLOOP4	☐	☒	☐	☐	☐	☐	N		00	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	129	1	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	130	2	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	131	3	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	132	4	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	133	5	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	134	6	I		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.
☐	135	7	I		☐	☐	☐	☐		☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.

AddDeleteMoveClear Used ByMove to/from backplane

S.I. = See Special Instructions

Manual Map Optimisation

Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Inputs and Outputs

Inputs and Outputs

☐ Enable Signal Required Check boxes

☒ Manual Allocation

Port Number & Type

Port:

21

☐ Inputs

☐ Outputs

☒ Inputs & Outputs

Card Type & Address

Serial IO 24/16

Card Address 6

	DET No	Bit No	Type I or O	Name	Req'd	BP	Count	Inv	U/D	Misc	DFM	DFM Group	Ext time	Phs	UTC	SDE	Pri	Used By HC	CC	IG	UD	LRT	Term Block	Line No
☐	168	0	O	LAMPSUP	☐	☐	☐	☐	☐	☐	N		00	☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	O-8
☐	169	1	O		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	O-9
☐	170	2	O		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	O-10
☐	171	3	O		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	O-11
☐	172	4	O		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	O-12
☐	173	5	O		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	O-13
☐	174	6	O		☐	☐	☐	☐	☐	☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	O-14
☐	175	7	O		☐	☐	☐	☐		☐				☐	☐	☐	☐	☐	☐	☐	☐	☐	S.I.	O-15

AddDeleteMoveClear Used ByMove to/from backplane

S.I. = See Special Instructions

Manual Map Optimisation

Last Modified 07/11/2023, Issue 1.12.16

Form Ref: 7.2 (7)

Works Order :
EM Number : 1084
Engineer :
Intersection : Lower Bristol Road / Oak Street Puffin

Aspect Drives (ELV Controllers)

Aspect Drives (ELV Controllers)

☒ Card Reversed

HPU Connection

1

Aspect Drive Configuration for LSLS 1 of 1 cards (Cabinet 1)

Output	Phase	Aspect	Use	Output	Phase	Aspect	Use
1	A	Red	Phase	17	N/A	N/A	N/A
2	A	Red	Phase	18	N/A	N/A	N/A
3	A	Red	Phase	19	N/A	N/A	N/A
4	A	Red	Phase	20	N/A	N/A	N/A
5	A	Red	Phase	21	N/A	N/A	N/A
6	A	Amber	Phase	22	N/A	N/A	N/A
7	A	Green	Phase	23	N/A	N/A	N/A
8	B	Red	Phase	24	N/A	N/A	N/A
9	B	Red	Phase	25	N/A	N/A	N/A
10	B	Amber	Phase	26	N/A	N/A	N/A
11	B	Amber	Phase	27	N/A	N/A	N/A
12	B	Green	Phase	28	N/A	N/A	N/A
13	B	Green	Phase	29	N/A	N/A	N/A
14	B	Green	Phase	30	N/A	N/A	N/A
15	B	Green	Phase	31	N/A	N/A	N/A
16	B	Green	Phase	32	N/A	N/A	N/A

I/O - DFM Group Timings

I/O - DFM Group Timings

Input Group	State	SET A	SET B	SET C	SET D
Group 0	Active (Mins)	30	30	30	30
	InActive (Hrs)	18	18	18	18
Group 1	Active (Mins)	30	30	30	30
	InActive (Hrs)	48	48	48	48
Group 2	Active (Mins)	10	10	10	10
	InActive (Hrs)	96	96	96	96
Group 3	Active (Mins)	30	30	30	30
	InActive (Hrs)	48	48	48	48
Group 4	Active (Mins)	30	30	30	30
	InActive (Hrs)	48	48	48	48
Group 5	Active (Mins)	10	10	10	10
	InActive (Hrs)	96	96	96	96
Group 6	Active (Mins)				
	InActive (Hrs)				
Group 7	Active (Mins)				
	InActive (Hrs)				

Note - 255 or blank disables DFM monitoring of that state (active or inactive) during that timeset (A to D)

Handset Limiting Values

State	Min	Max
Active (Mins)	0	255
InActive (Hrs)	0	255

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